

BFS Traversal

Step 1: Start at A

Legend

Complete

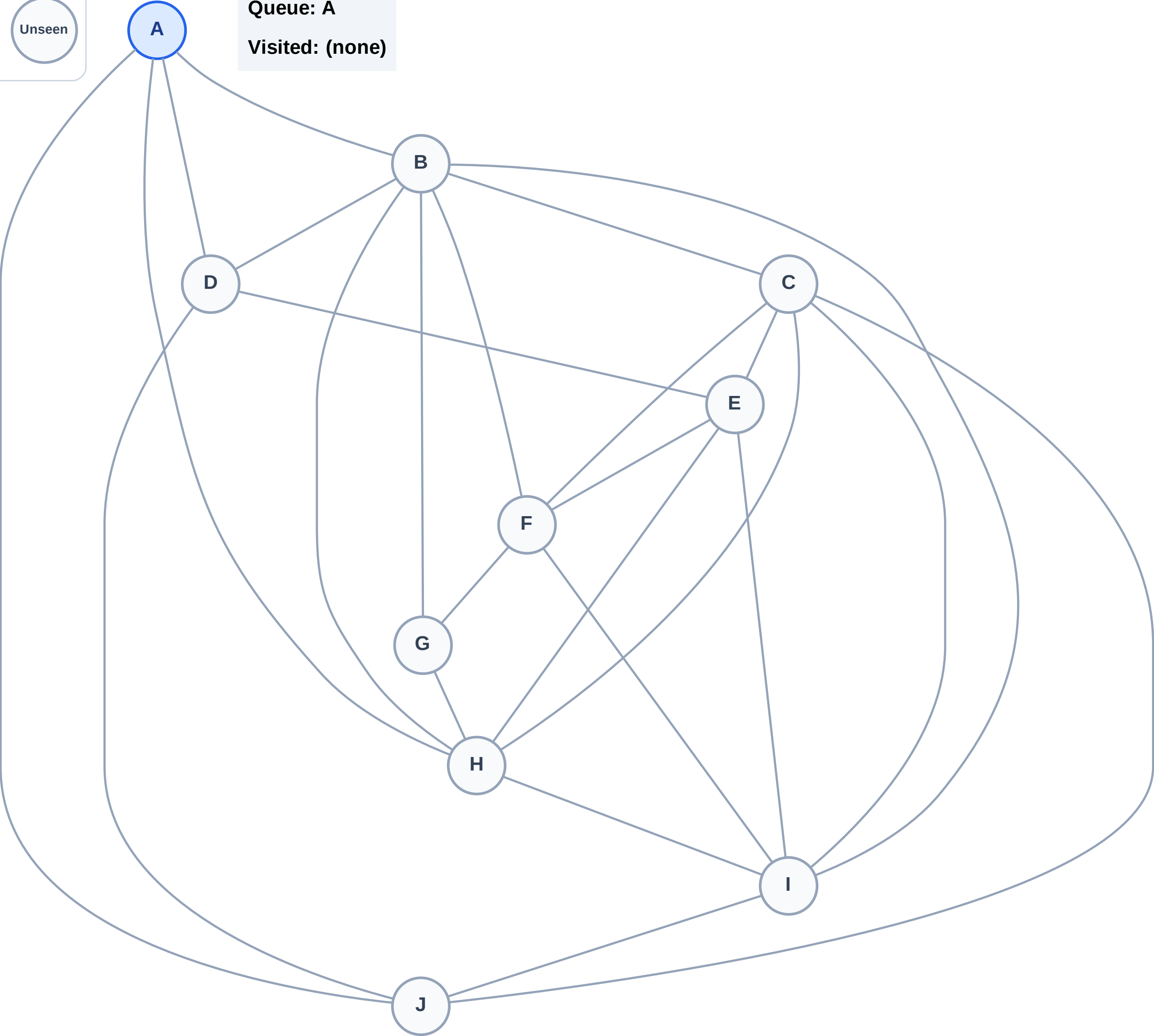
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

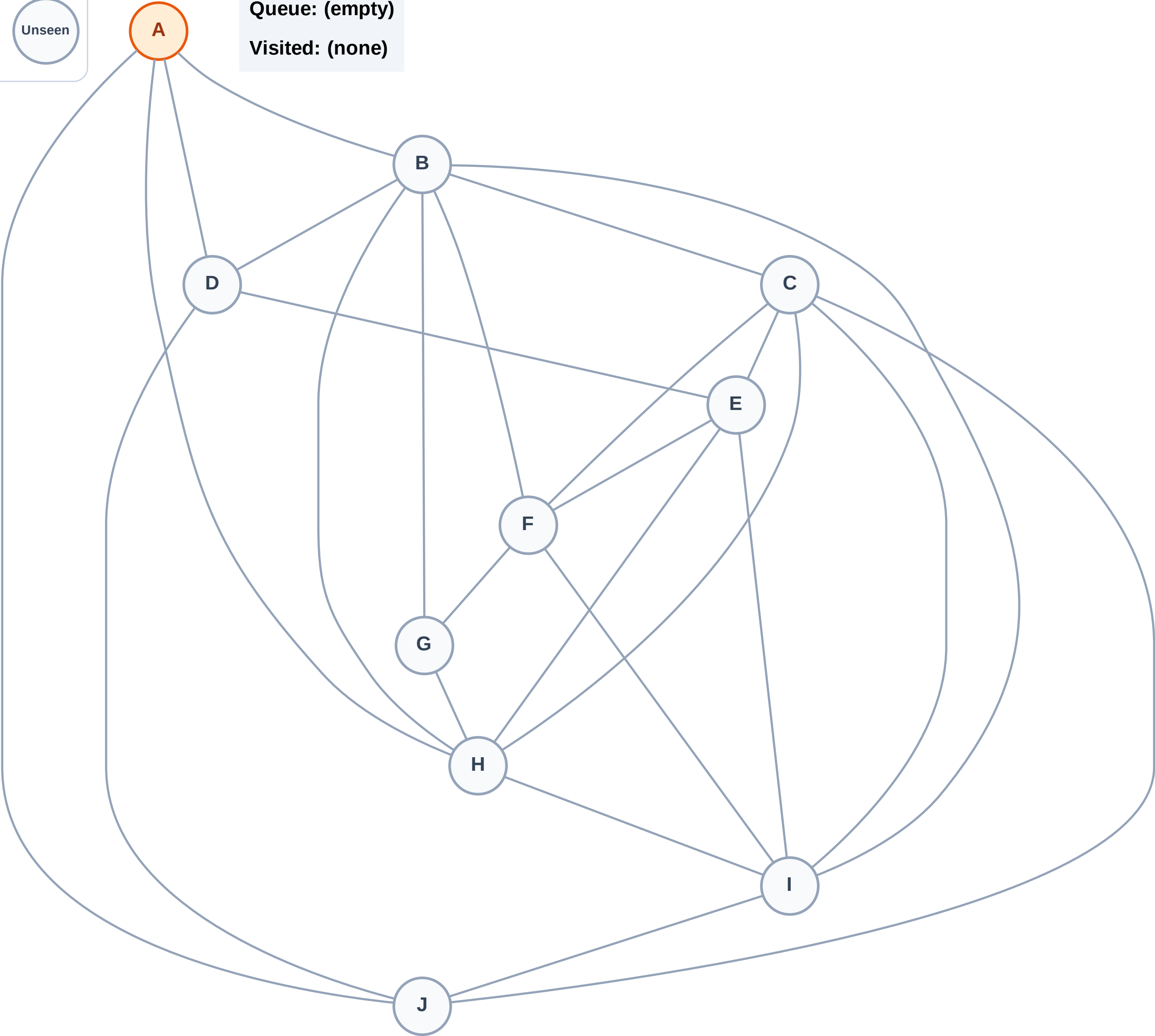
Complete

Processing

Discovered

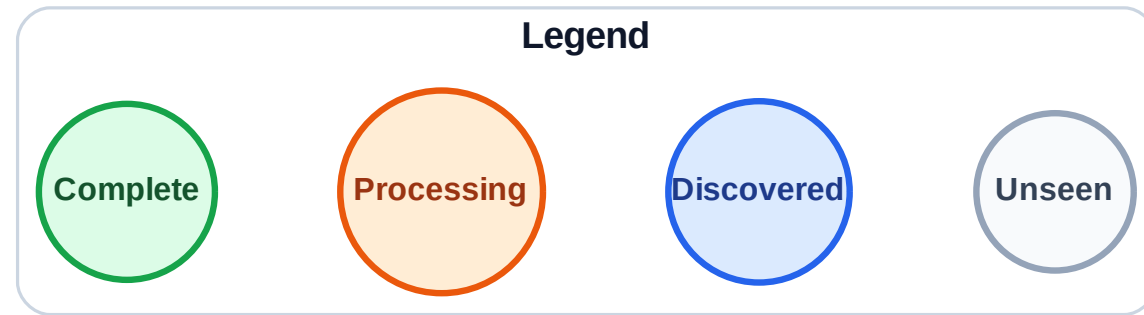
Unseen

Queue: (empty)
Visited: (none)

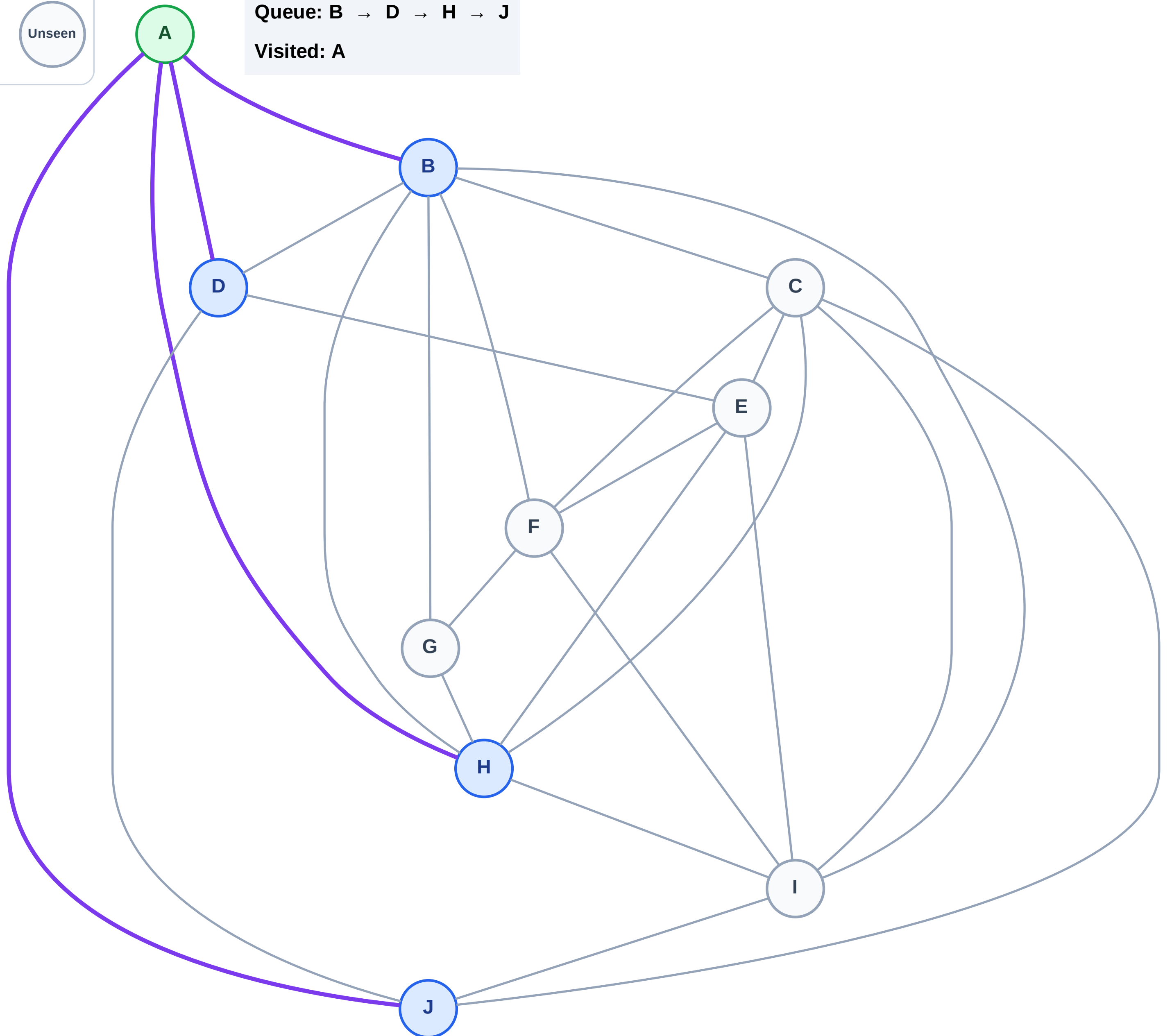


BFS Traversal

Step 3: Discover B, D, H, J from A



Queue: B → D → H → J
Visited: A



BFS Traversal

Step 4: Process node B

Legend

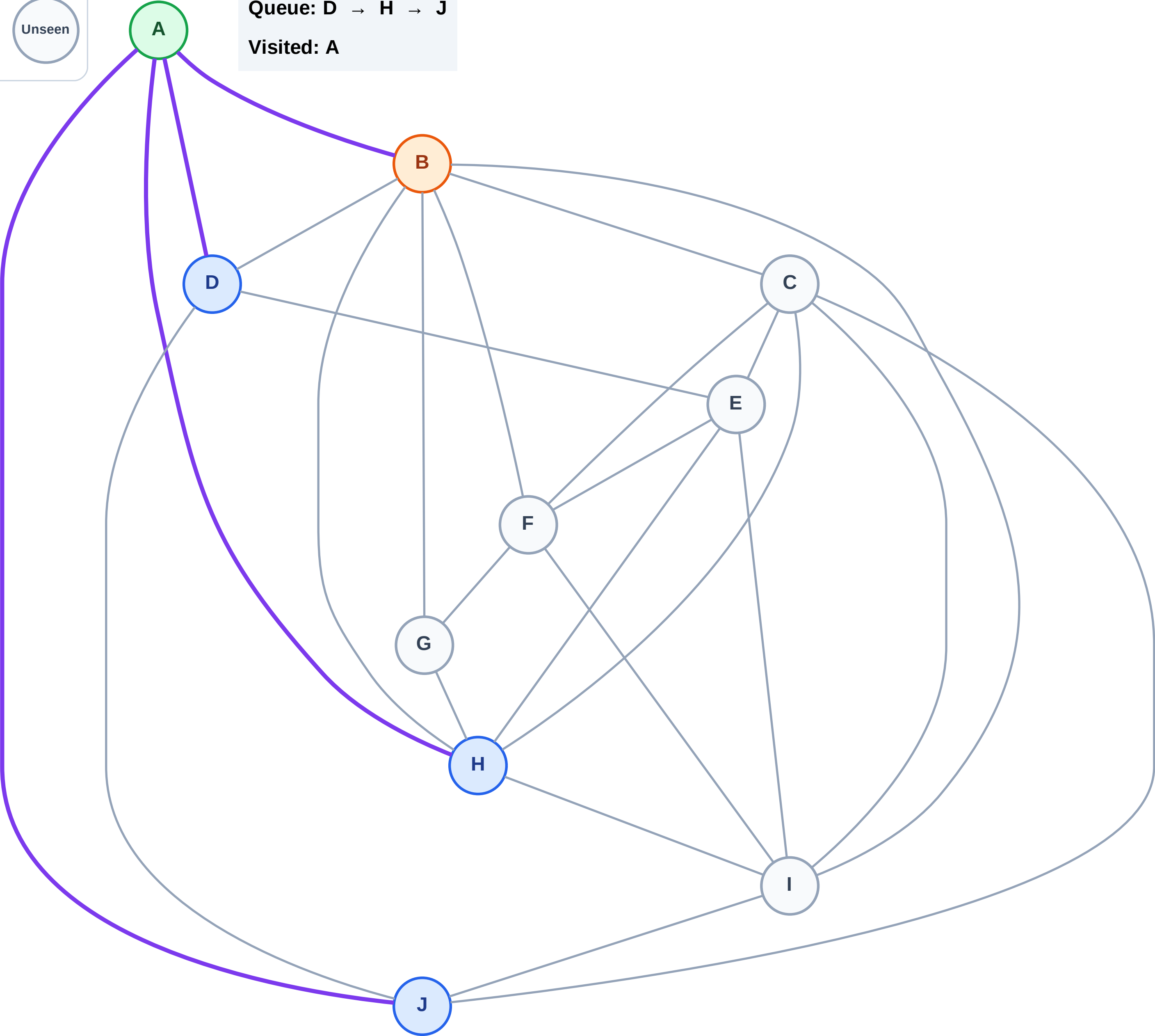
Complete

Processing

Discovered

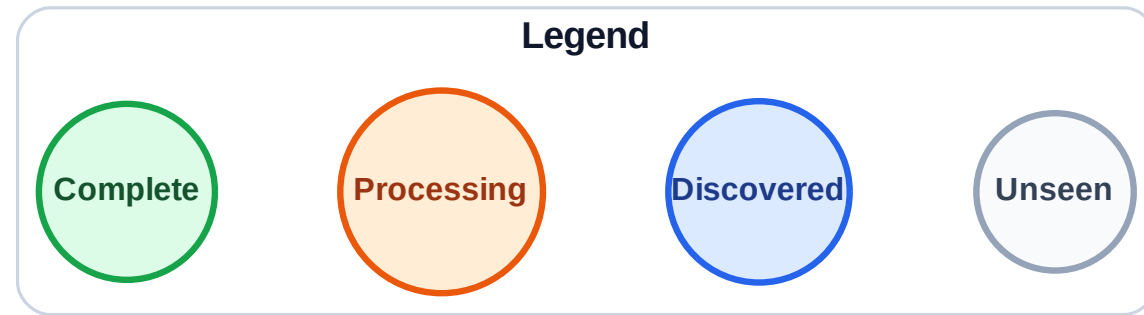
Unseen

Queue: D → H → J
Visited: A

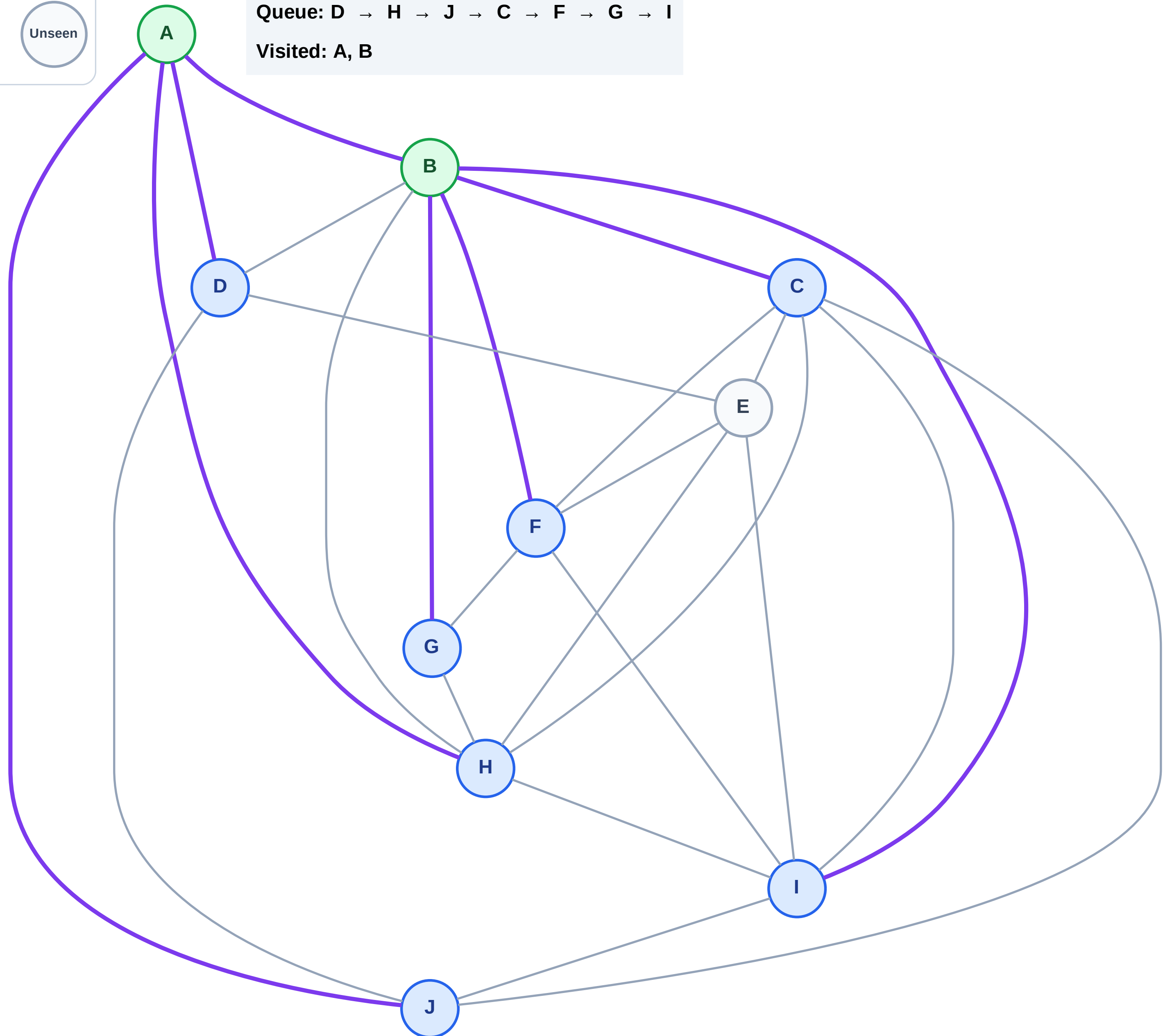


BFS Traversal

Step 5: Discover C, F, G, I from B



Queue: D → H → J → C → F → G → I
Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

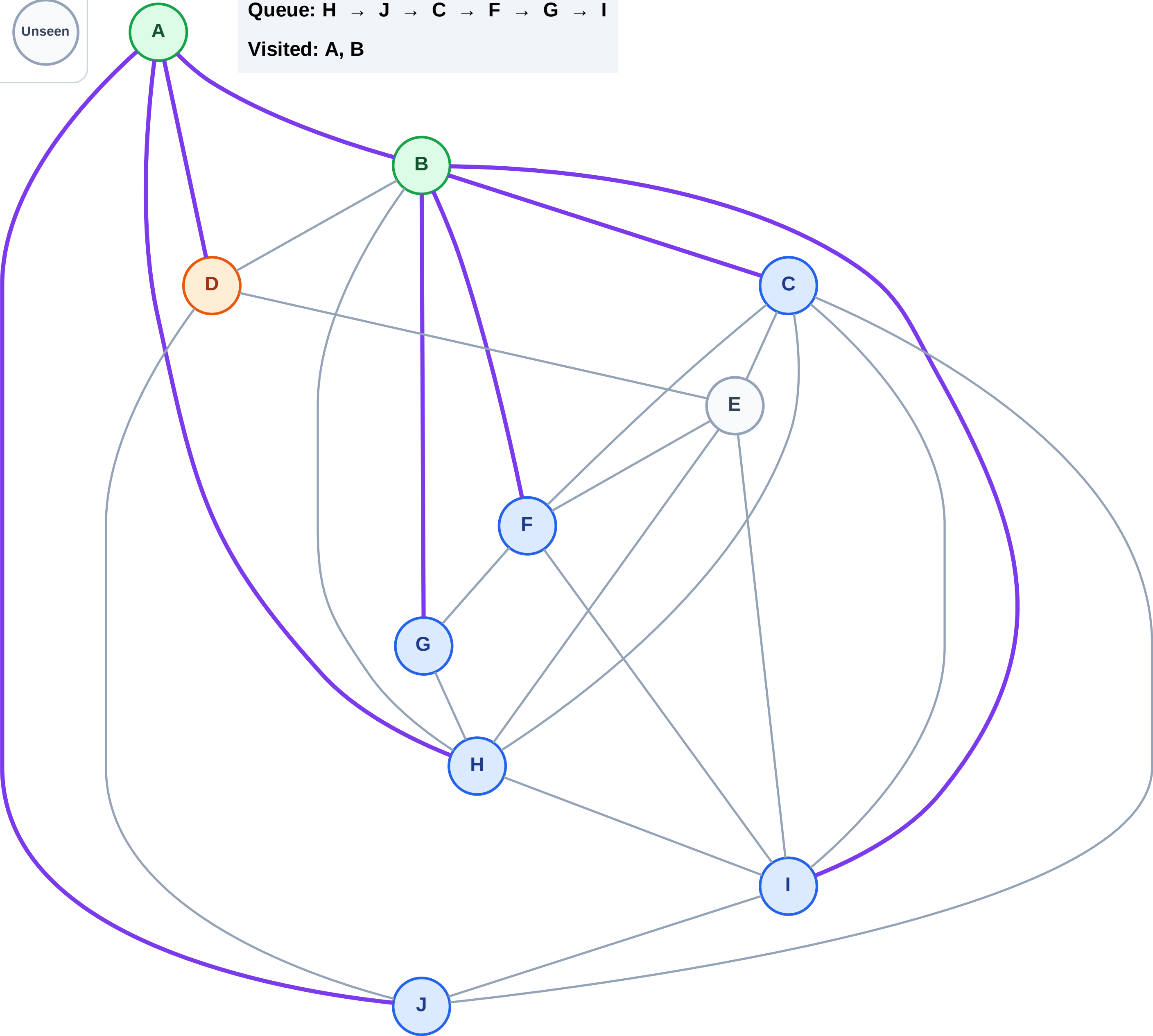
Processing

Discovered

Unseen

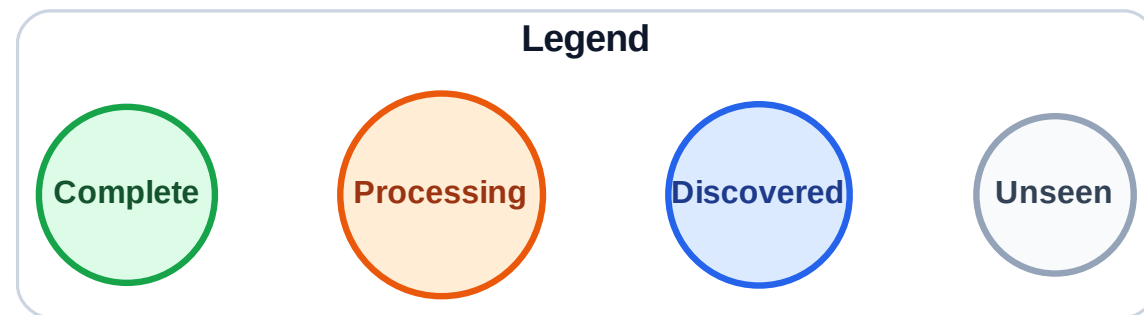
Queue: H → J → C → F → G → I

Visited: A, B

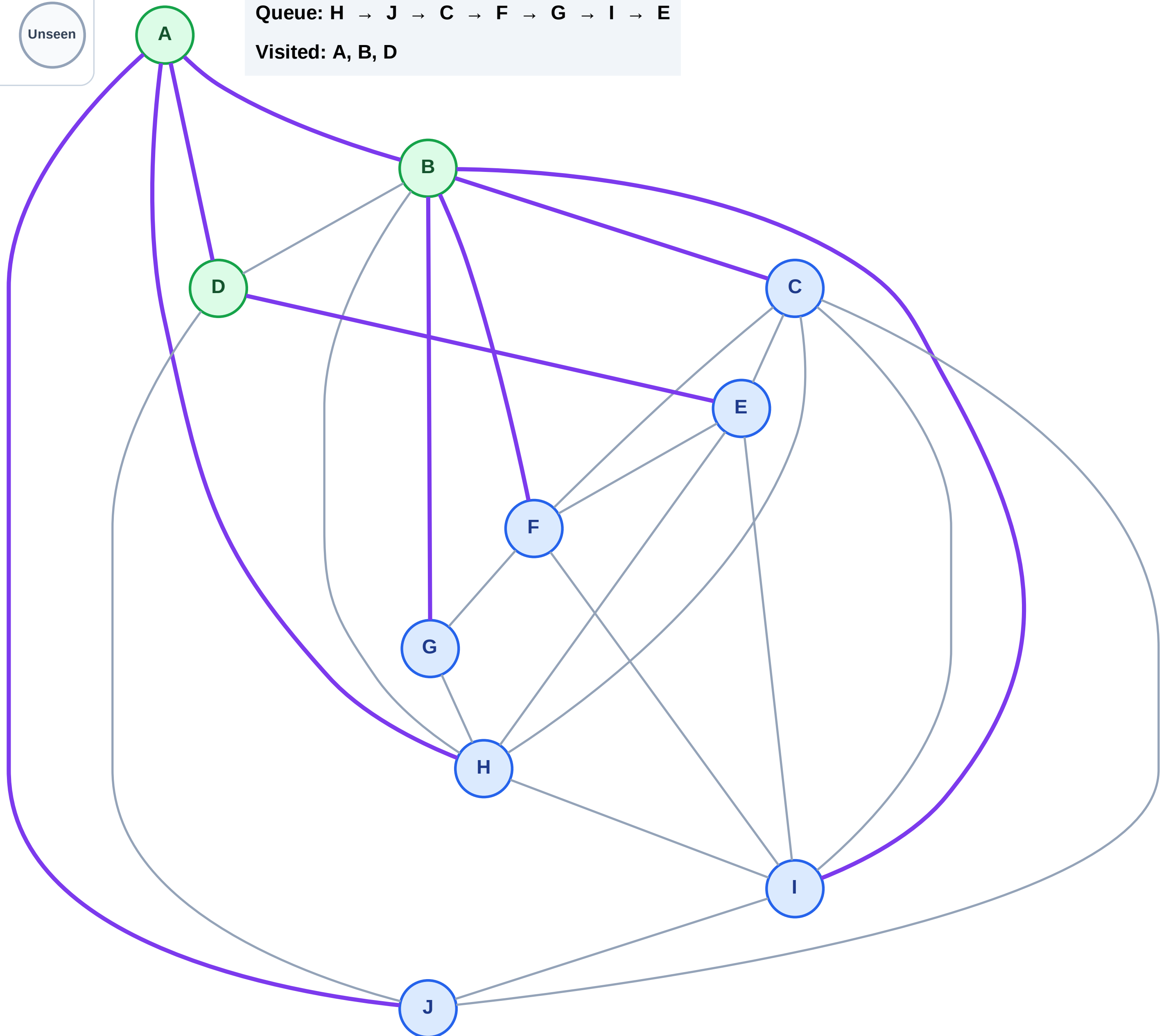


BFS Traversal

Step 7: Discover E from D



Queue: H → J → C → F → G → I → E
Visited: A, B, D



BFS Traversal

Step 8: Process node H

Legend

Complete

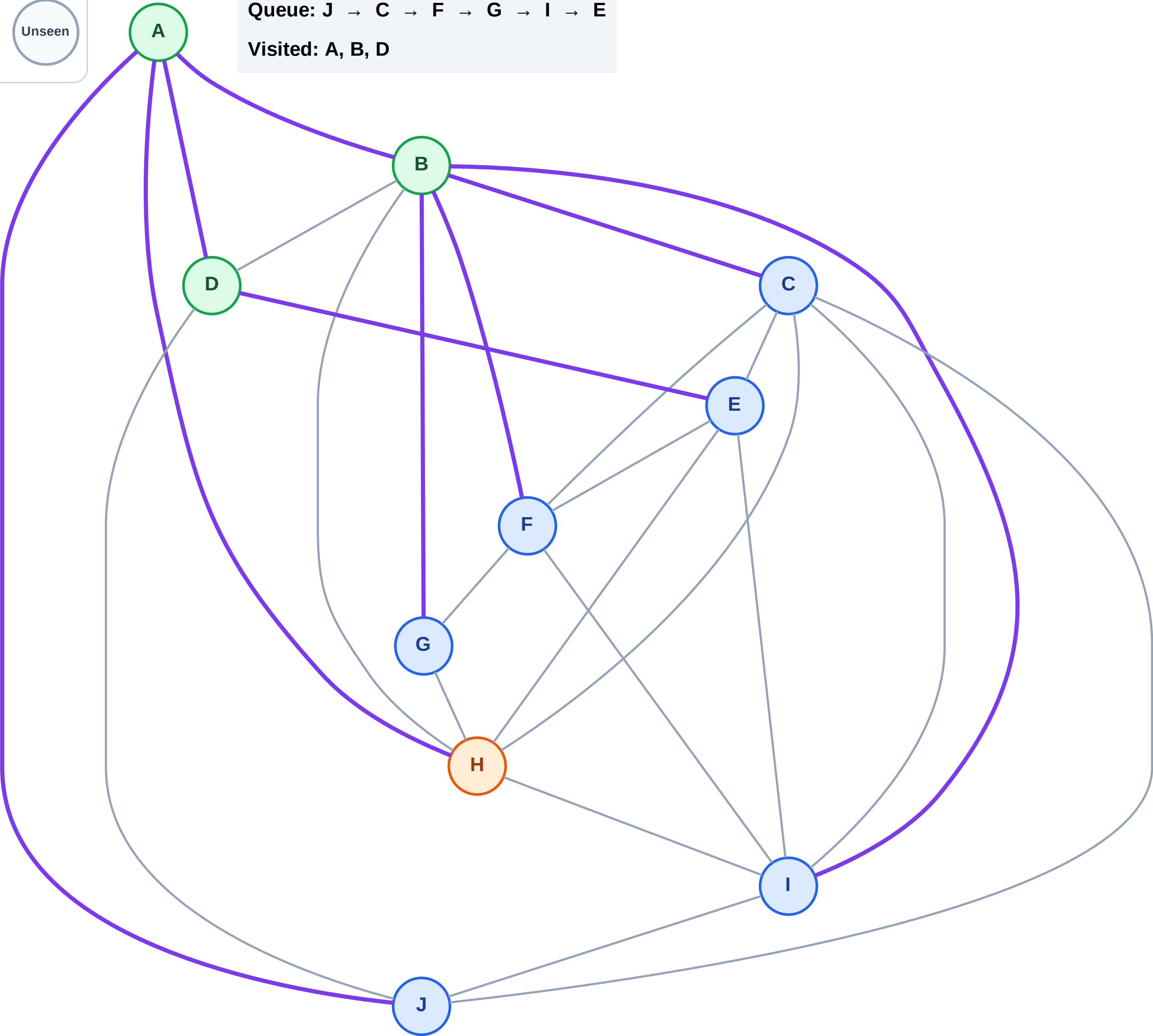
Processing

Discovered

Unseen

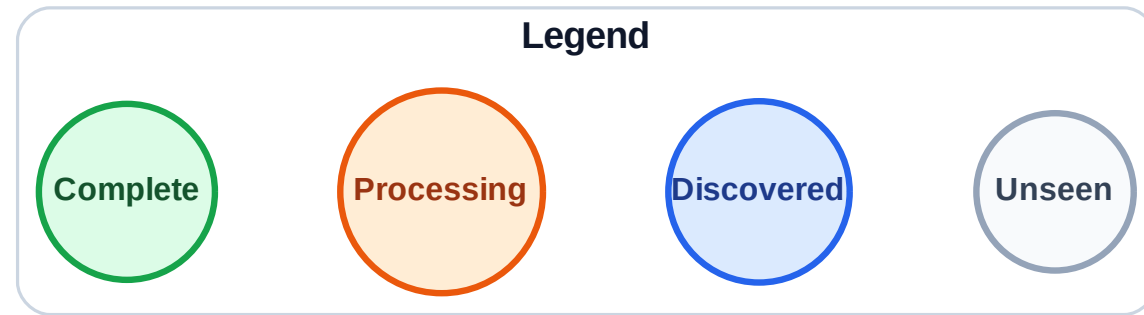
Queue: J → C → F → G → I → E

Visited: A, B, D

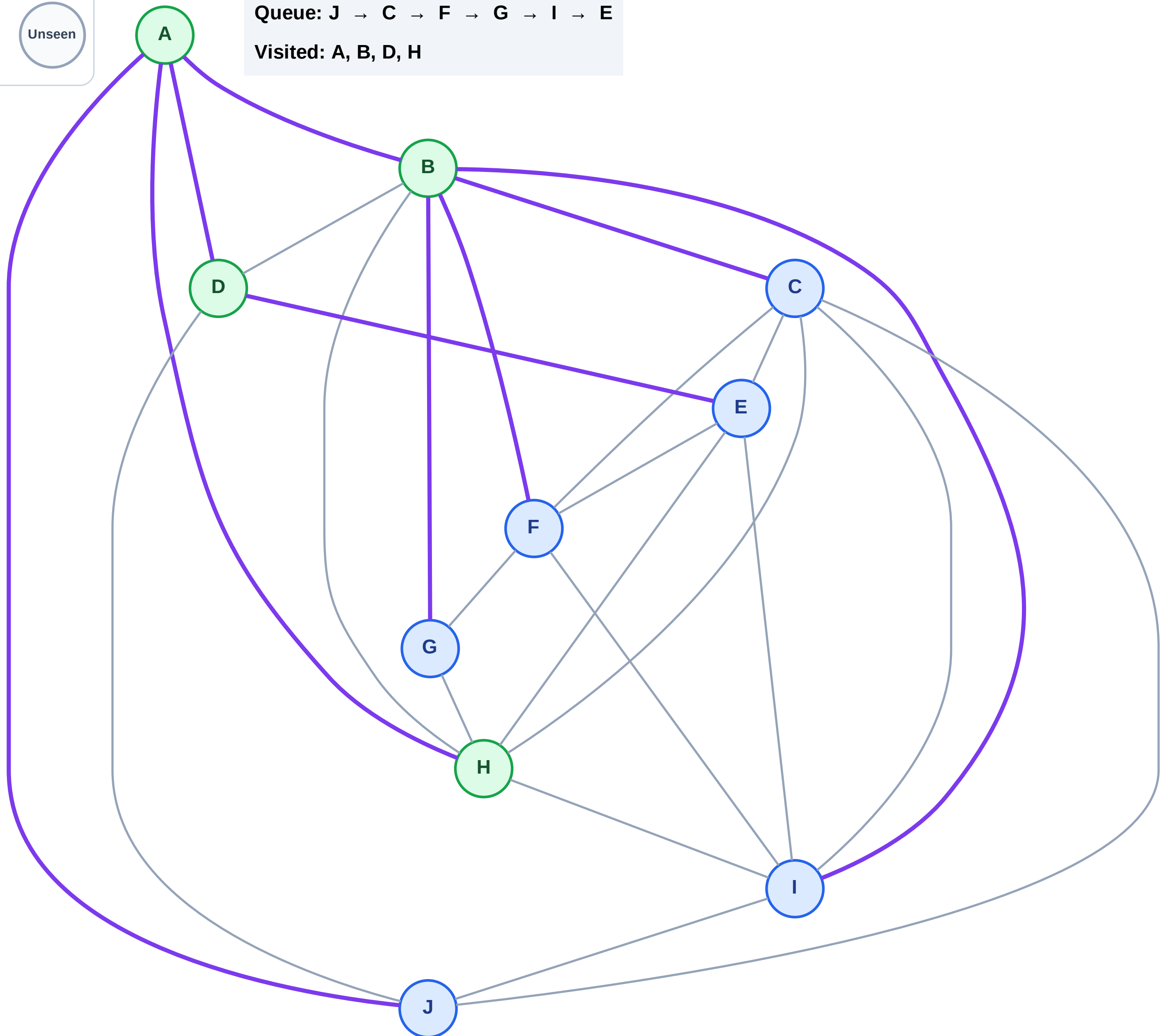


BFS Traversal

Step 9: No new neighbors from H

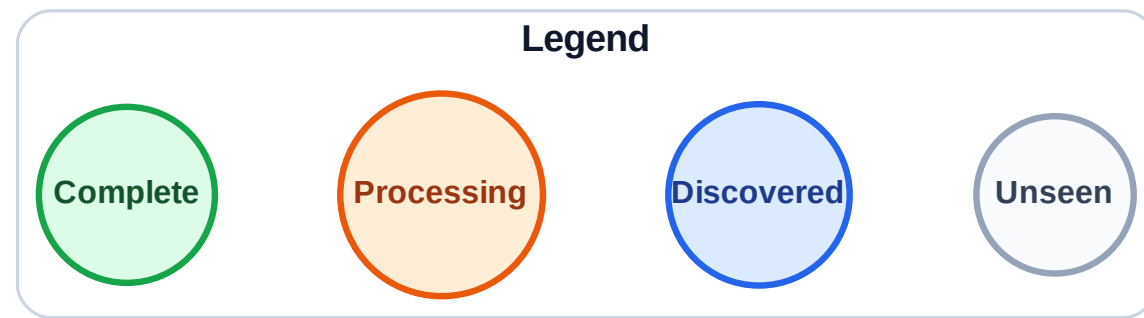


Queue: J → C → F → G → I → E
Visited: A, B, D, H



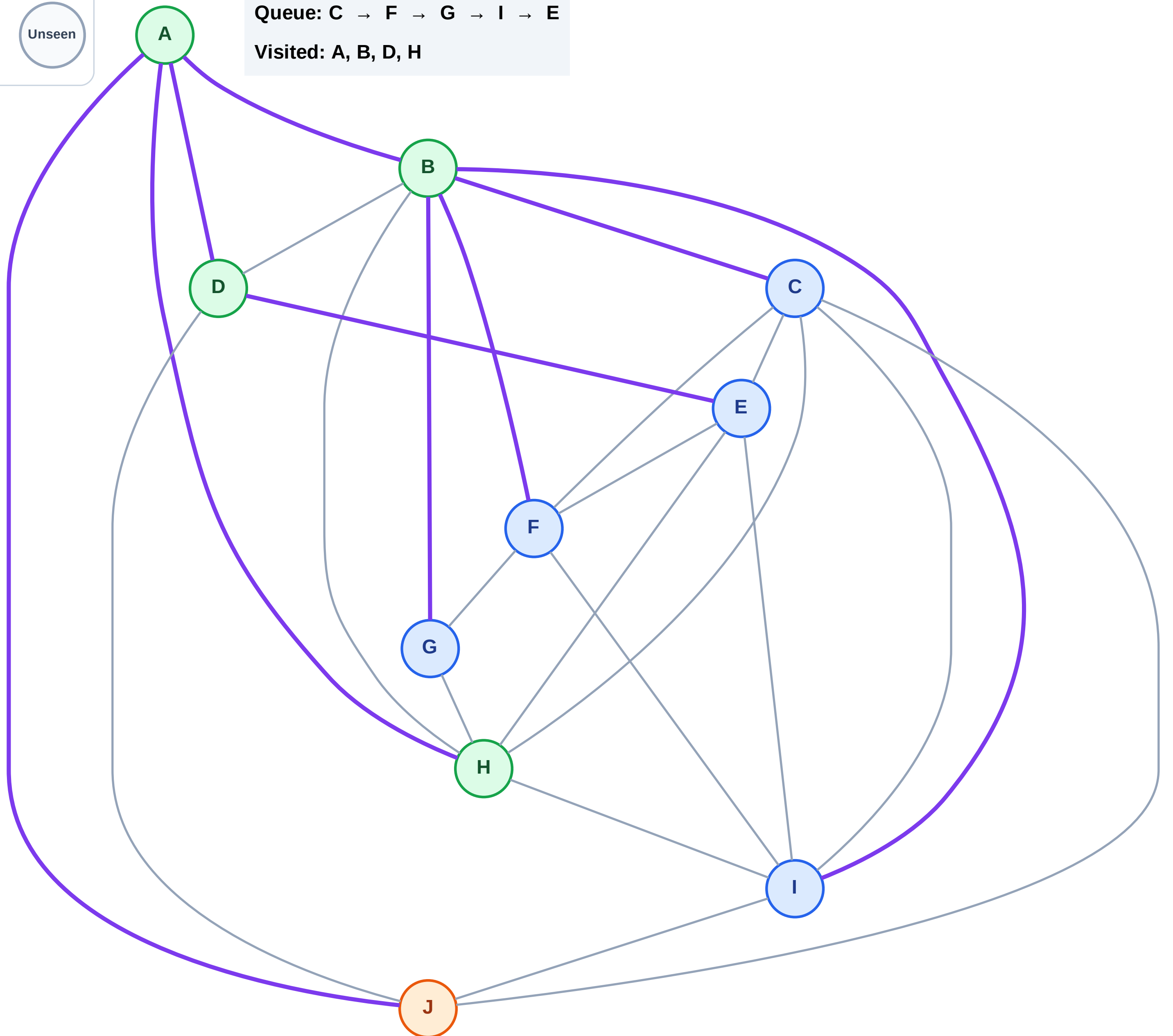
BFS Traversal

Step 10: Process node J



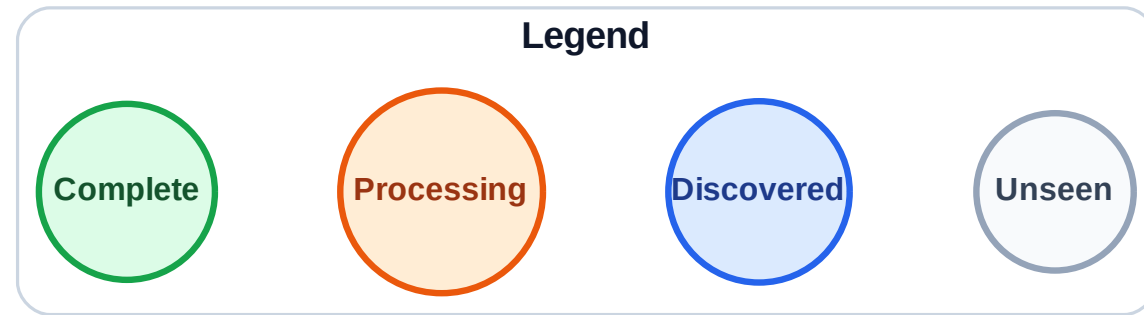
Queue: C → F → G → I → E

Visited: A, B, D, H

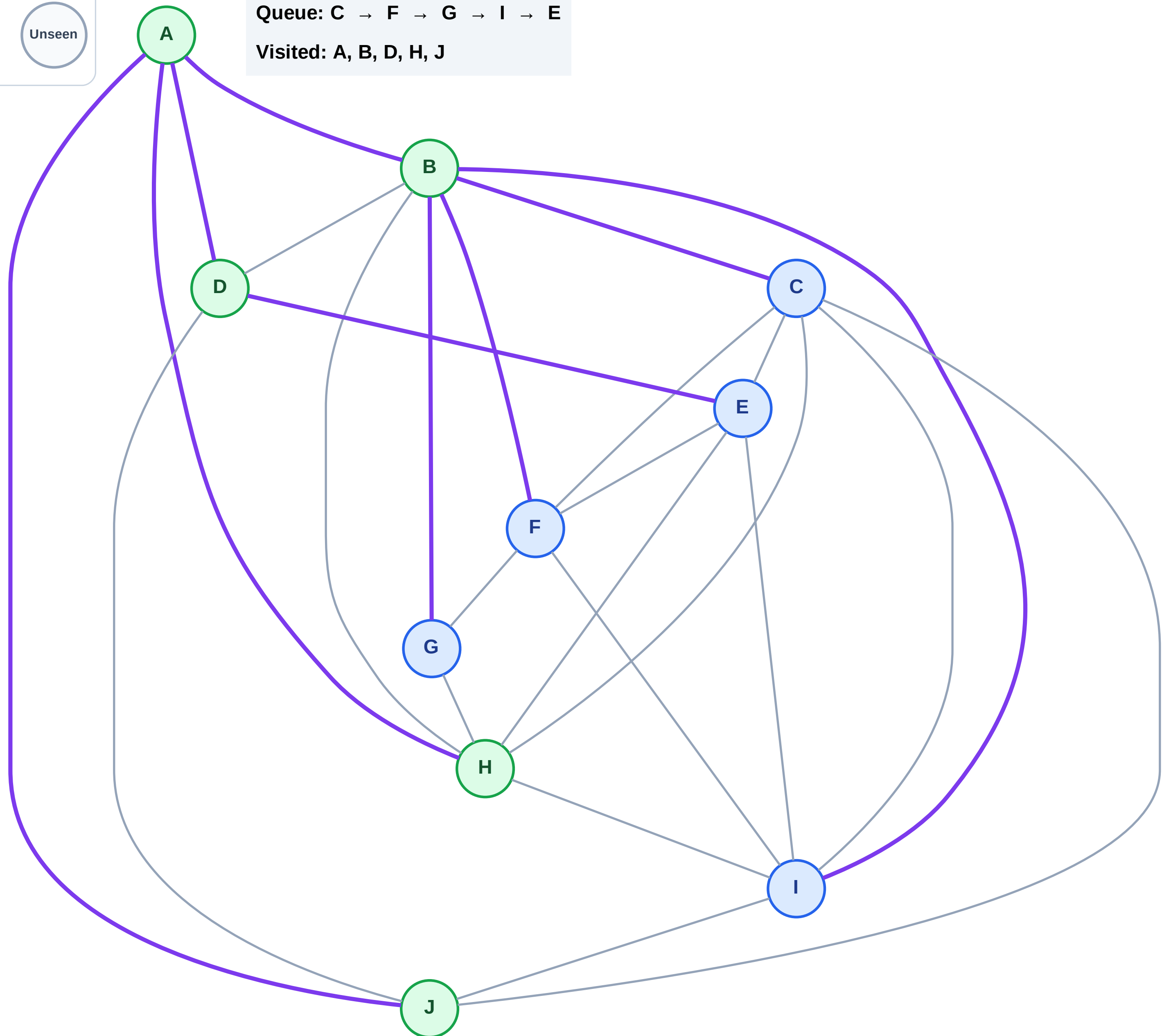


BFS Traversal

Step 11: No new neighbors from J

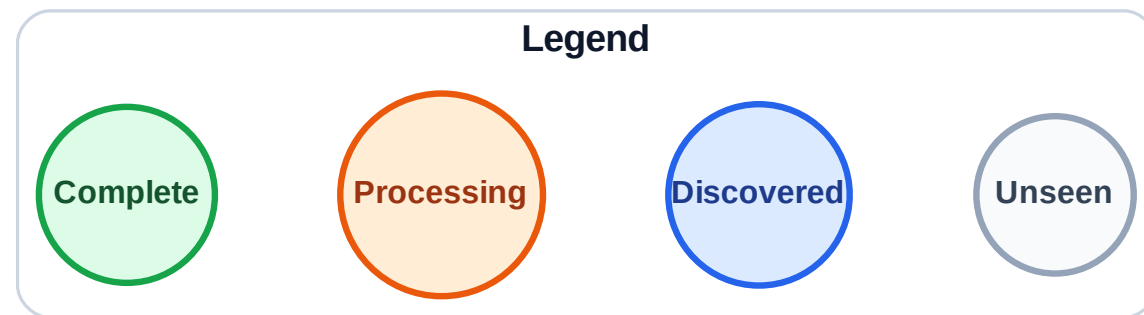


Queue: C → F → G → I → E
Visited: A, B, D, H, J



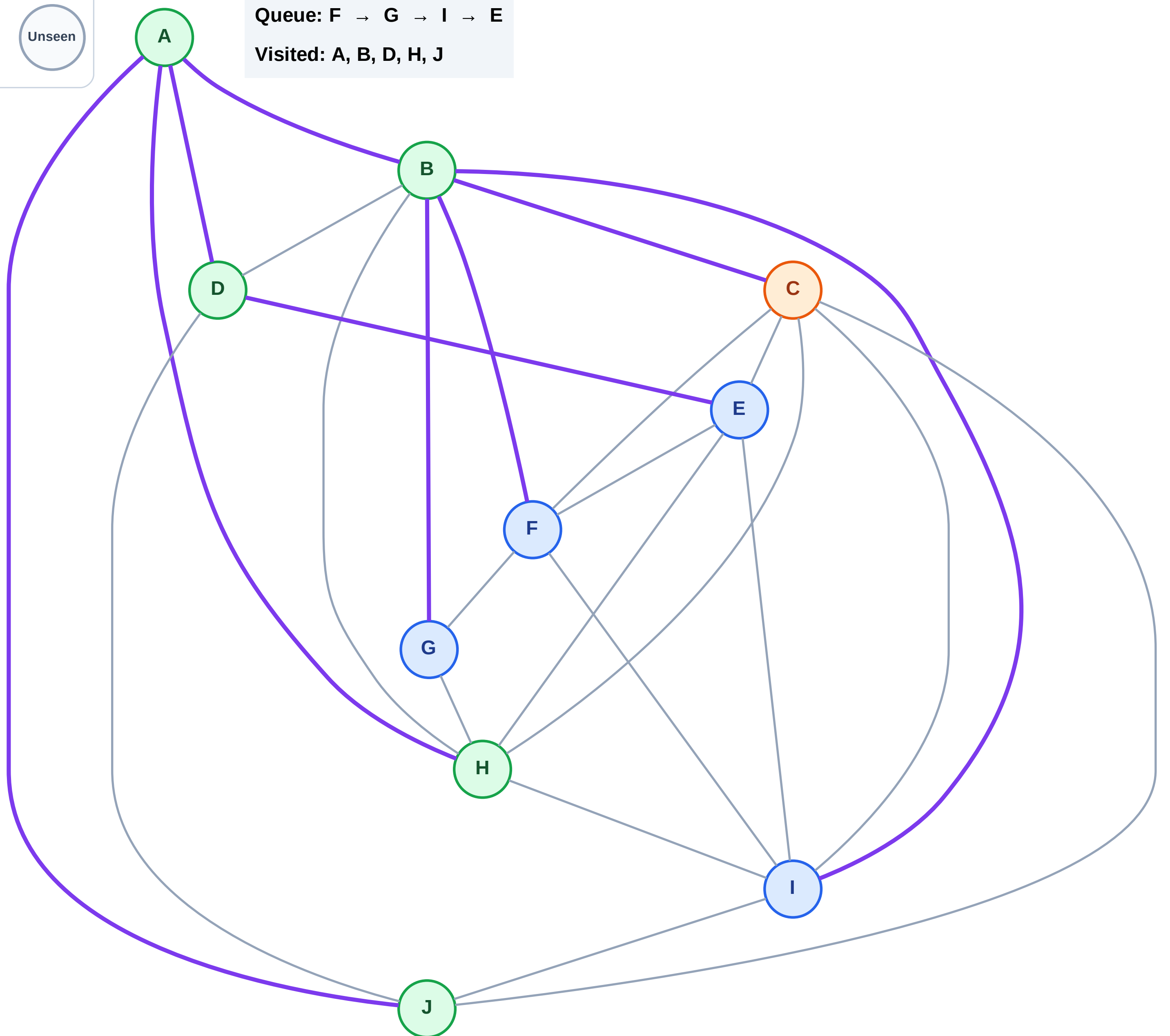
BFS Traversal

Step 12: Process node C



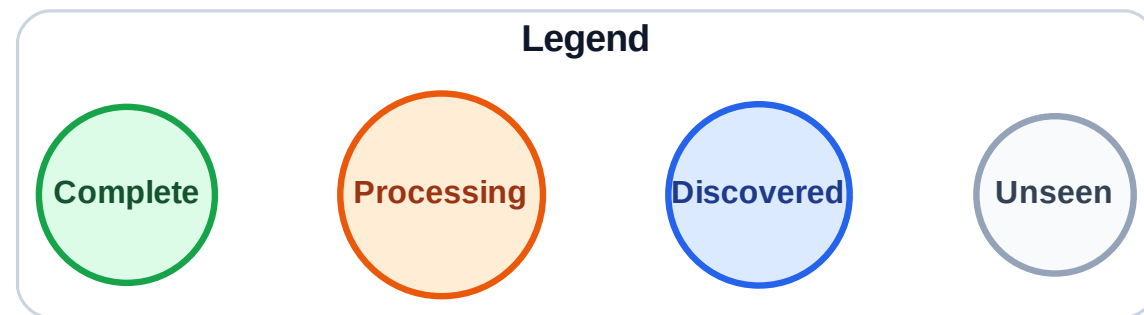
Queue: F → G → I → E

Visited: A, B, D, H, J



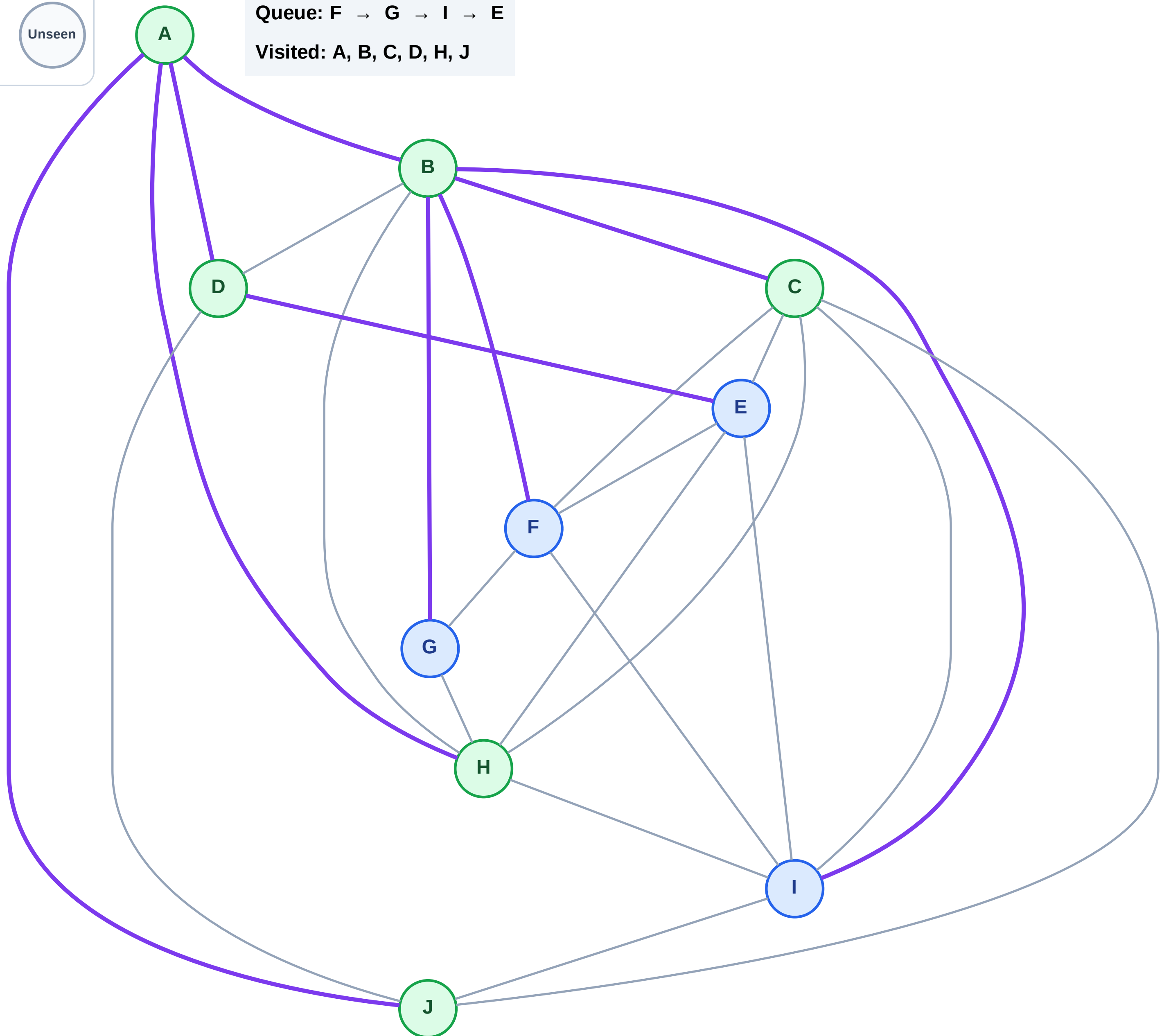
BFS Traversal

Step 13: No new neighbors from C



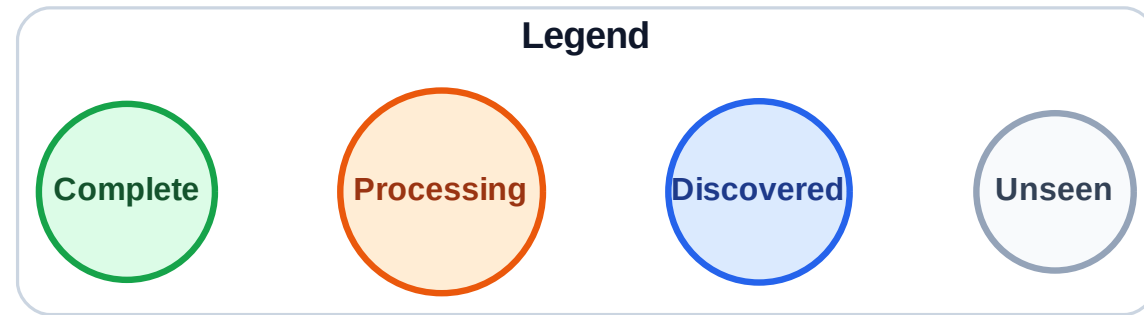
Queue: F → G → I → E

Visited: A, B, C, D, H, J

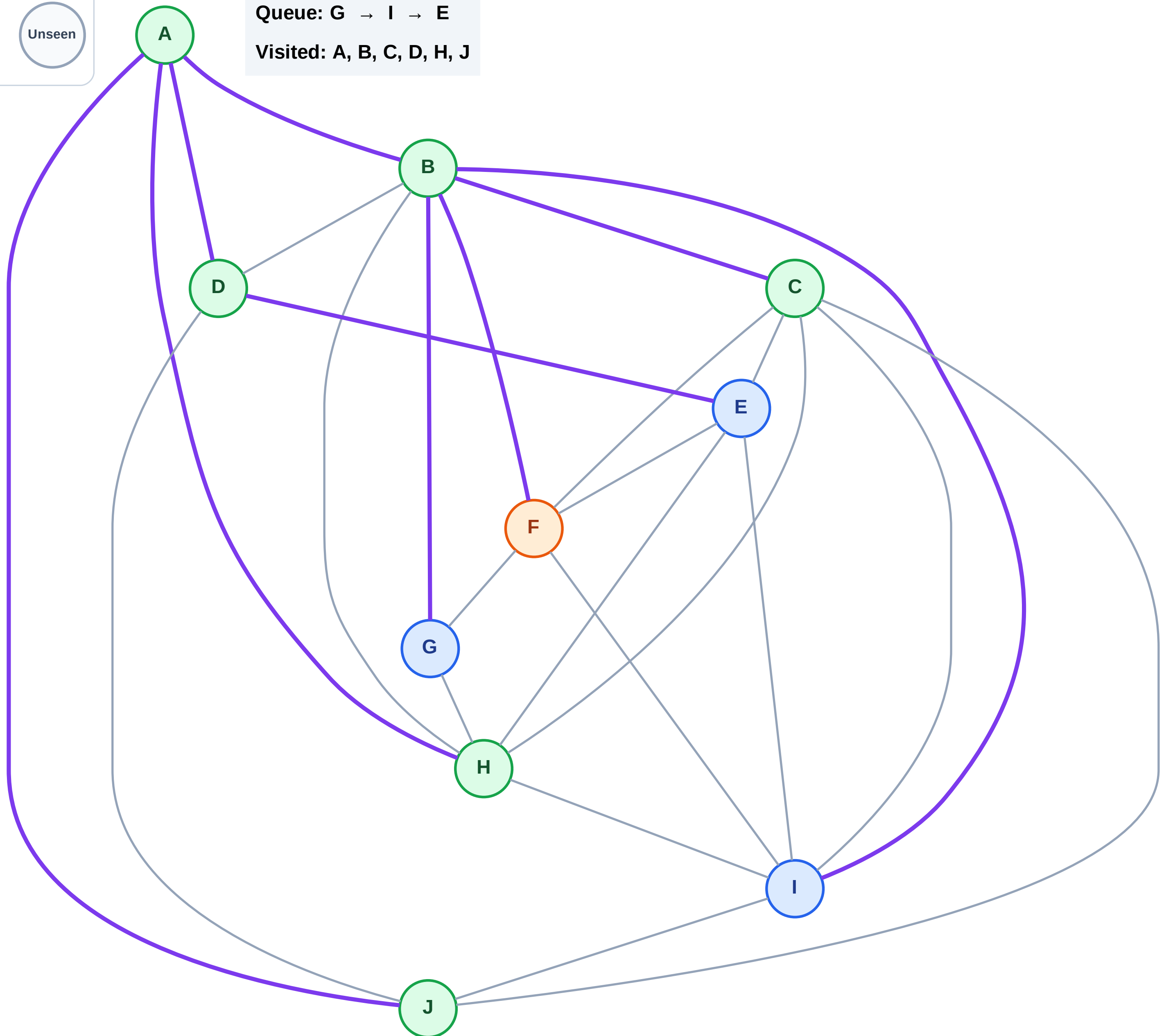


BFS Traversal

Step 14: Process node F



Queue: G → I → E
Visited: A, B, C, D, H, J

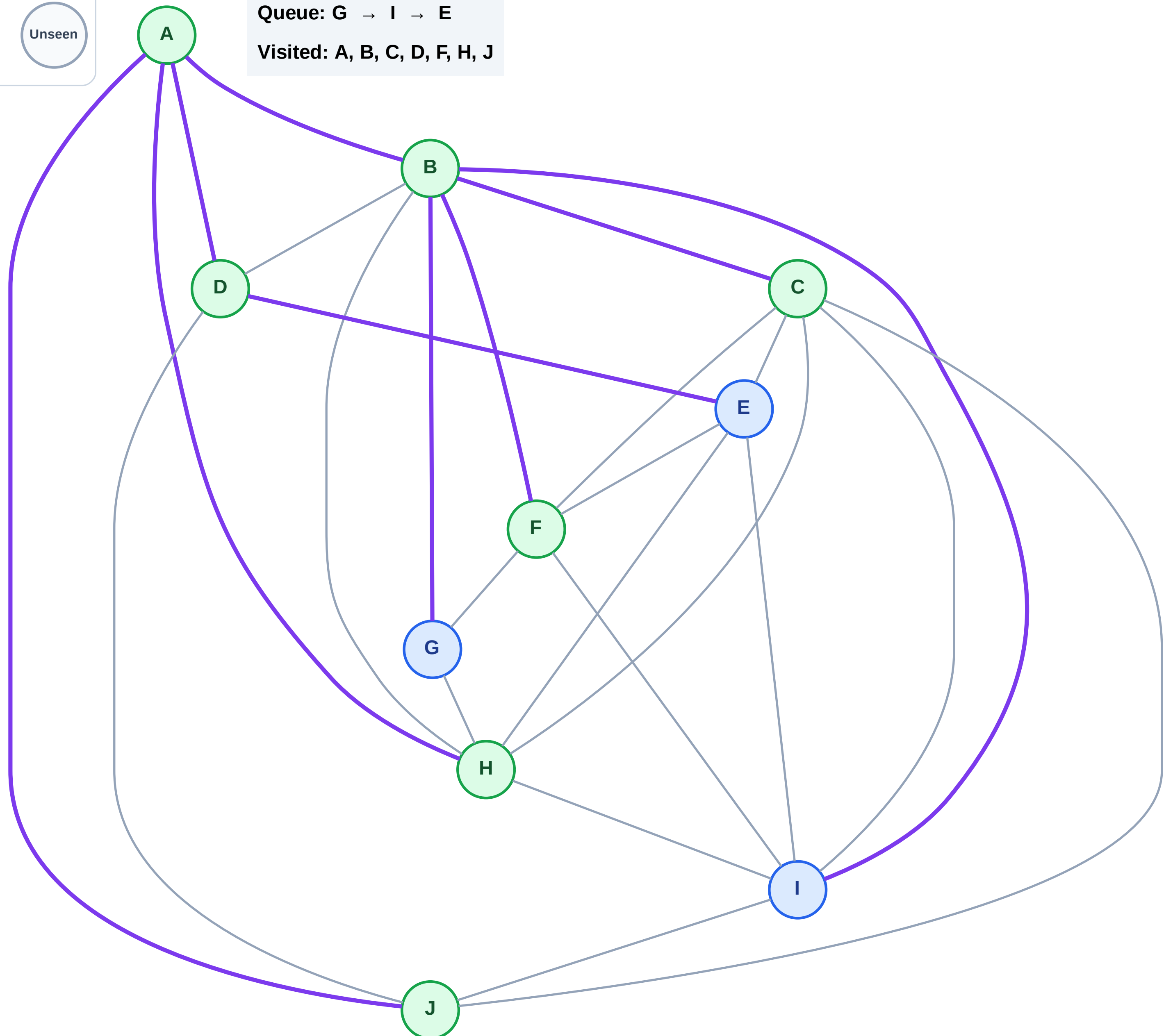
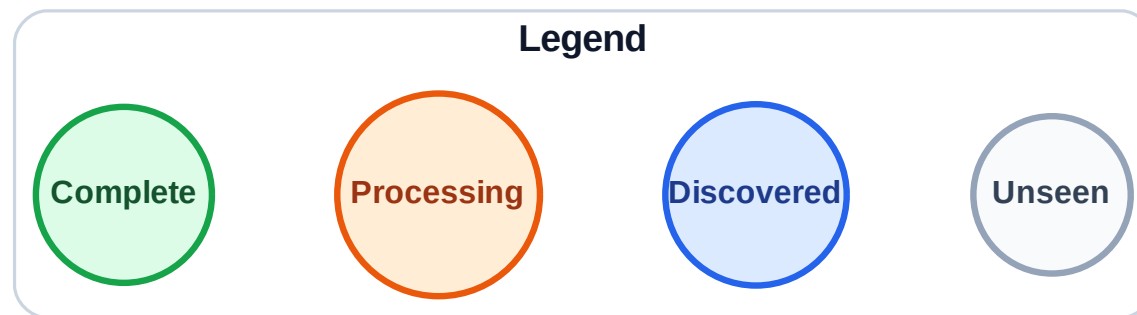


BFS Traversal

Step 15: No new neighbors from F

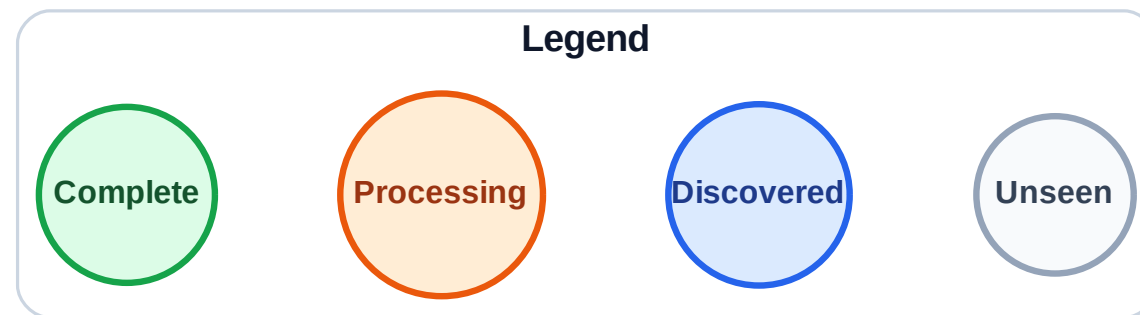
Queue: G → I → E

Visited: A, B, C, D, F, H, J



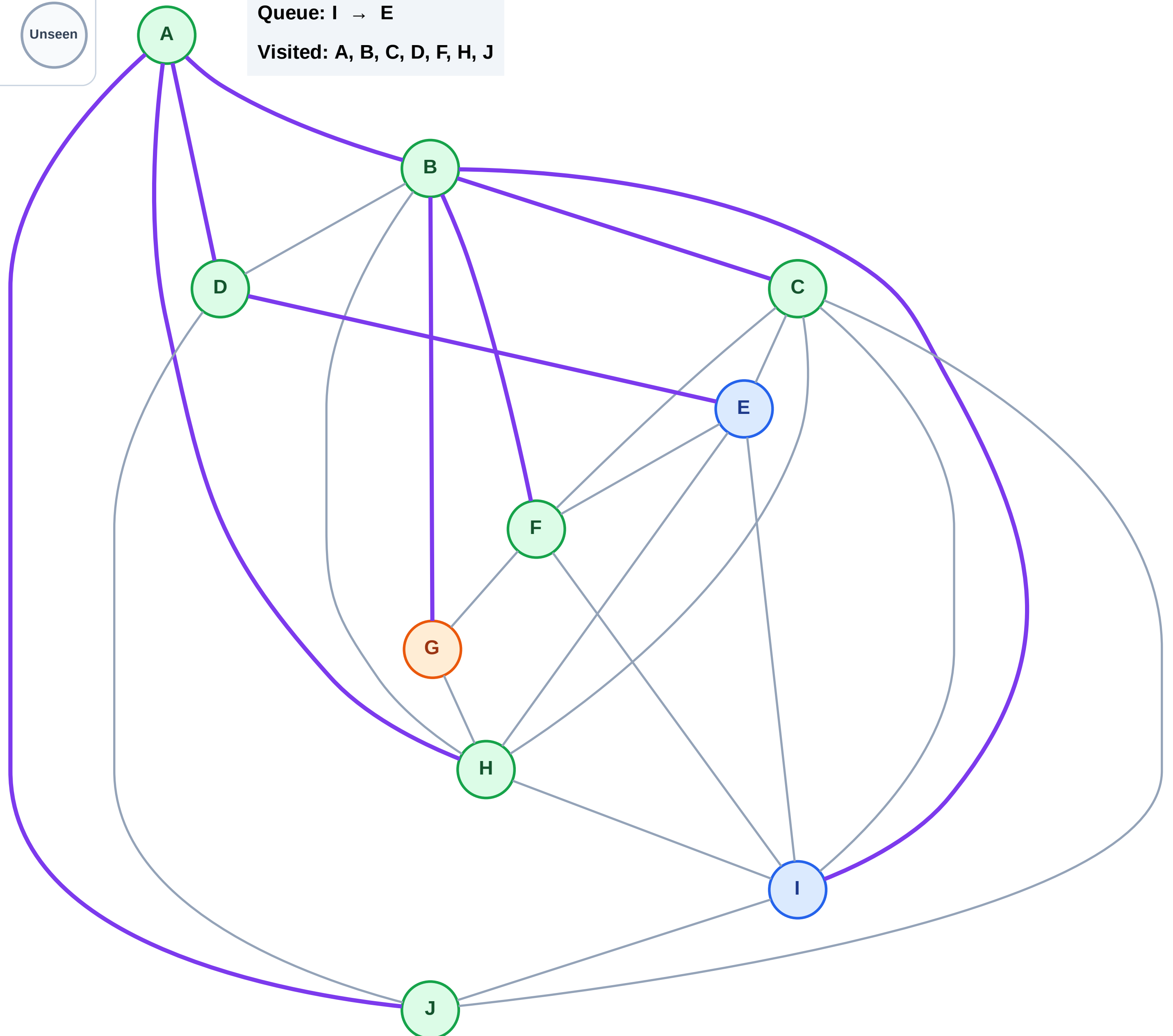
BFS Traversal

Step 16: Process node G



Queue: I → E

Visited: A, B, C, D, F, H, J

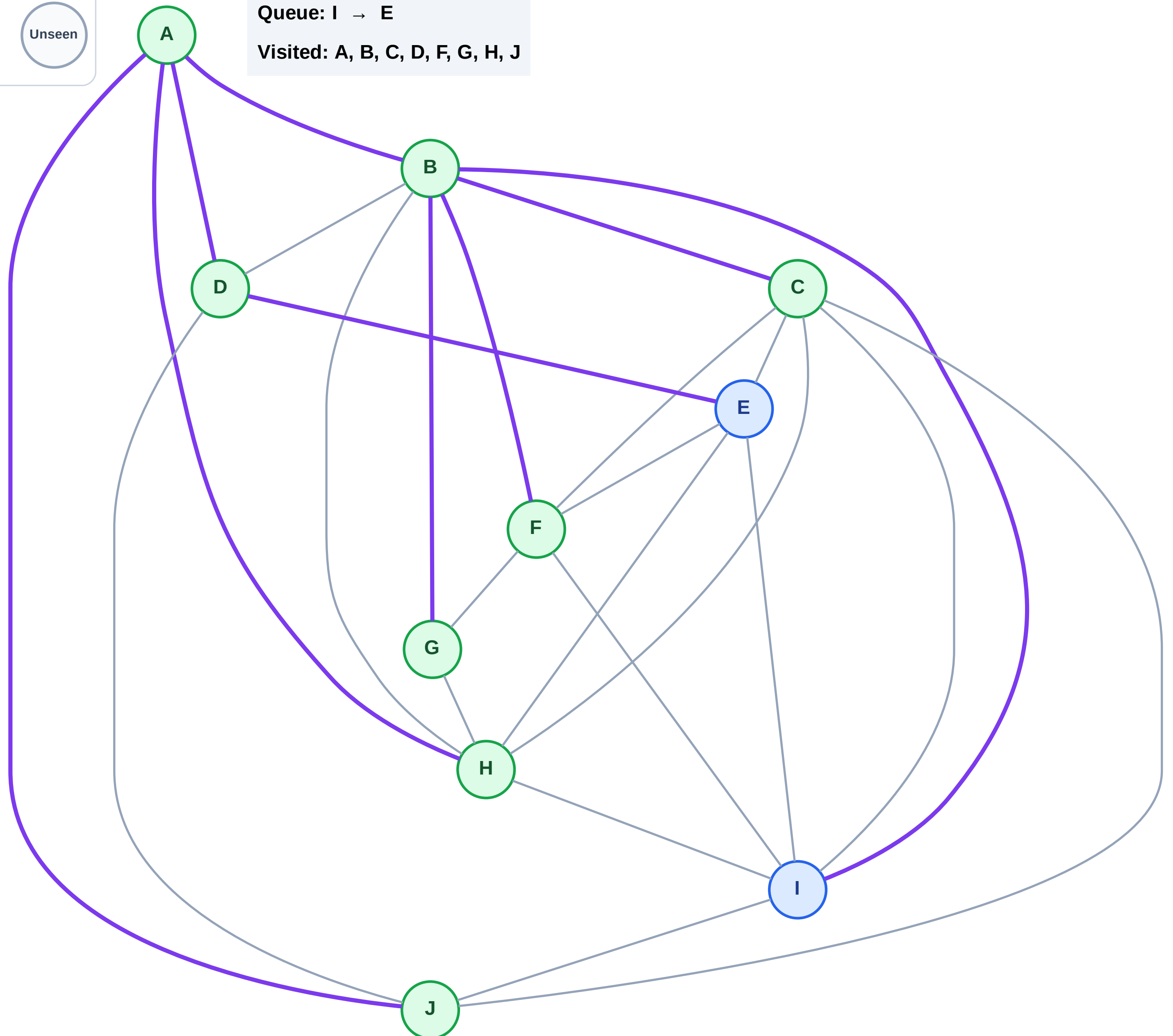
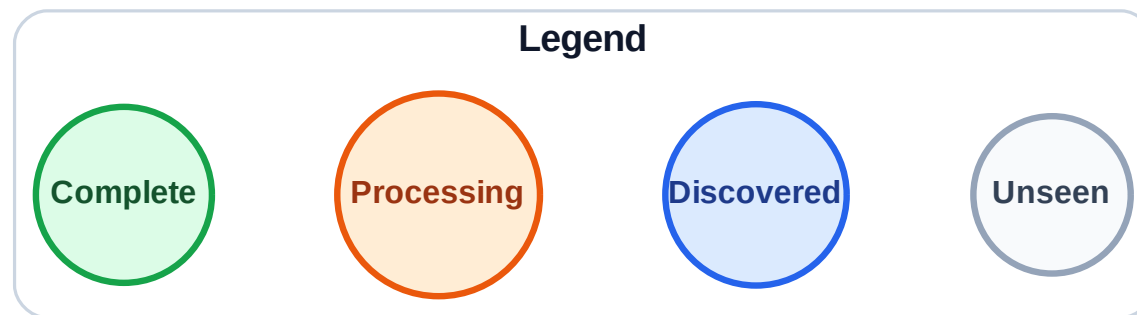


BFS Traversal

Step 17: No new neighbors from G

Queue: I → E

Visited: A, B, C, D, F, G, H, J



BFS Traversal

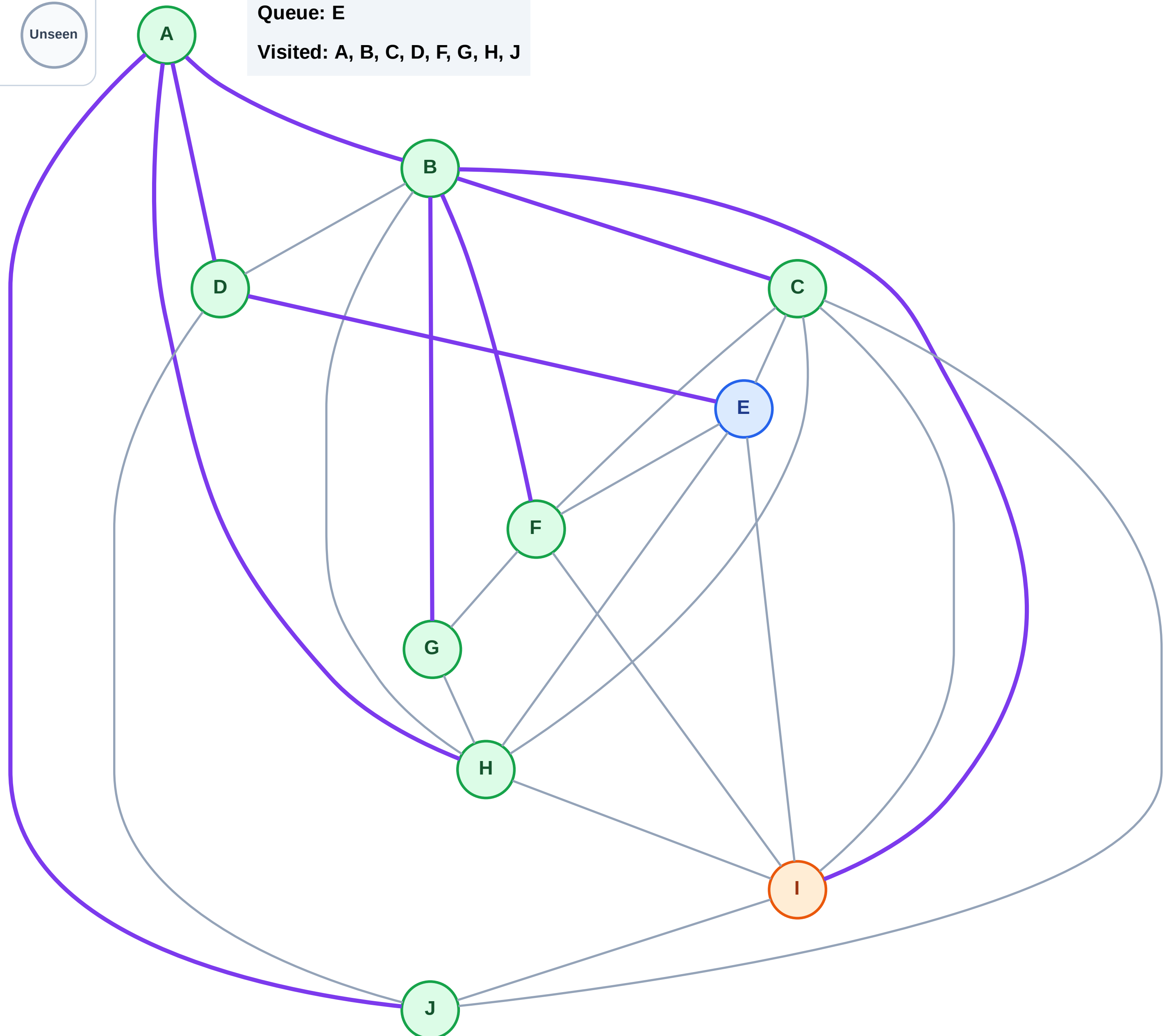
Step 18: Process node I

Legend



Queue: E

Visited: A, B, C, D, F, G, H, J



BFS Traversal

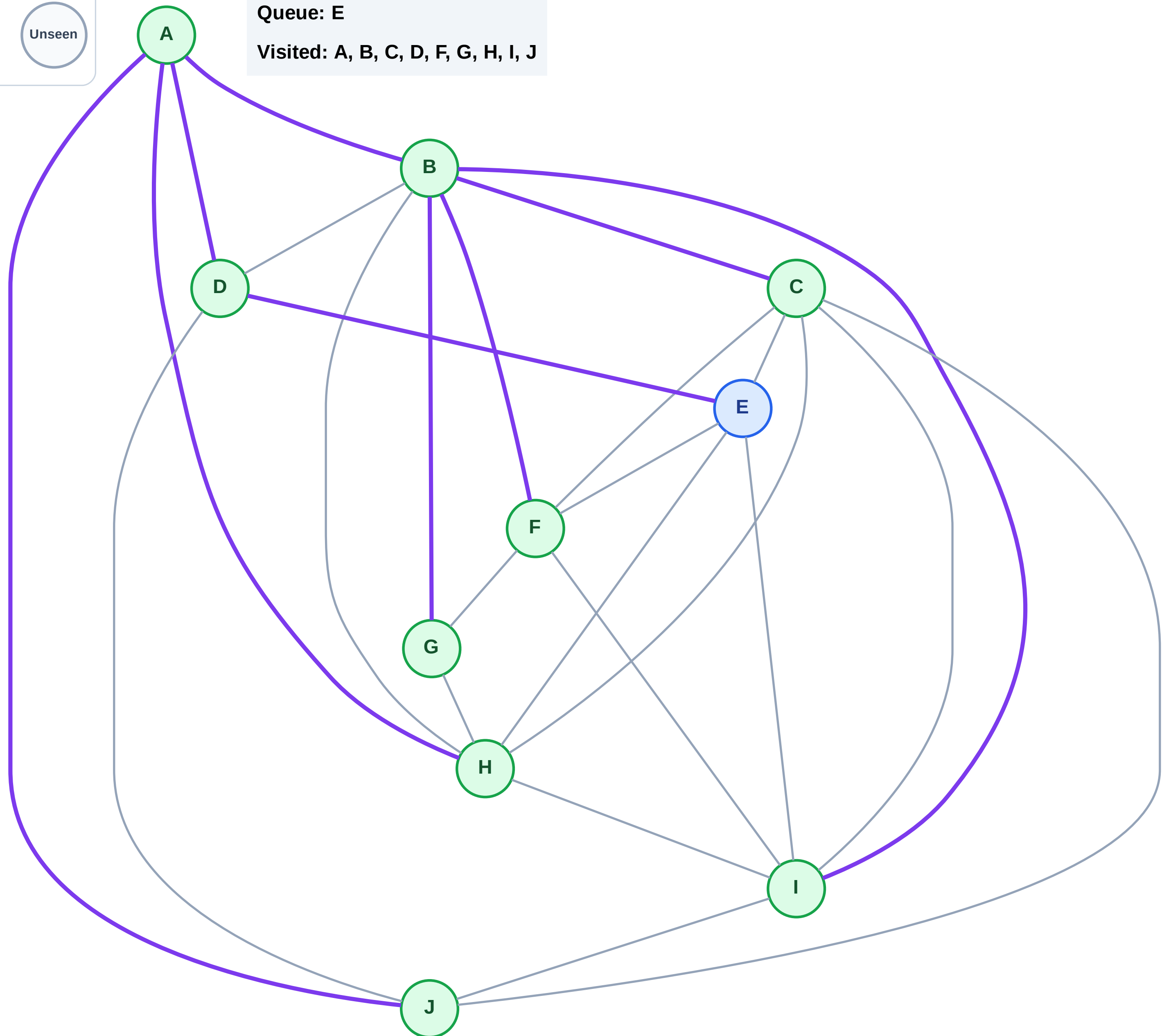
Step 19: No new neighbors from I

Legend



Queue: E

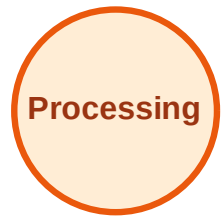
Visited: A, B, C, D, F, G, H, I, J



BFS Traversal

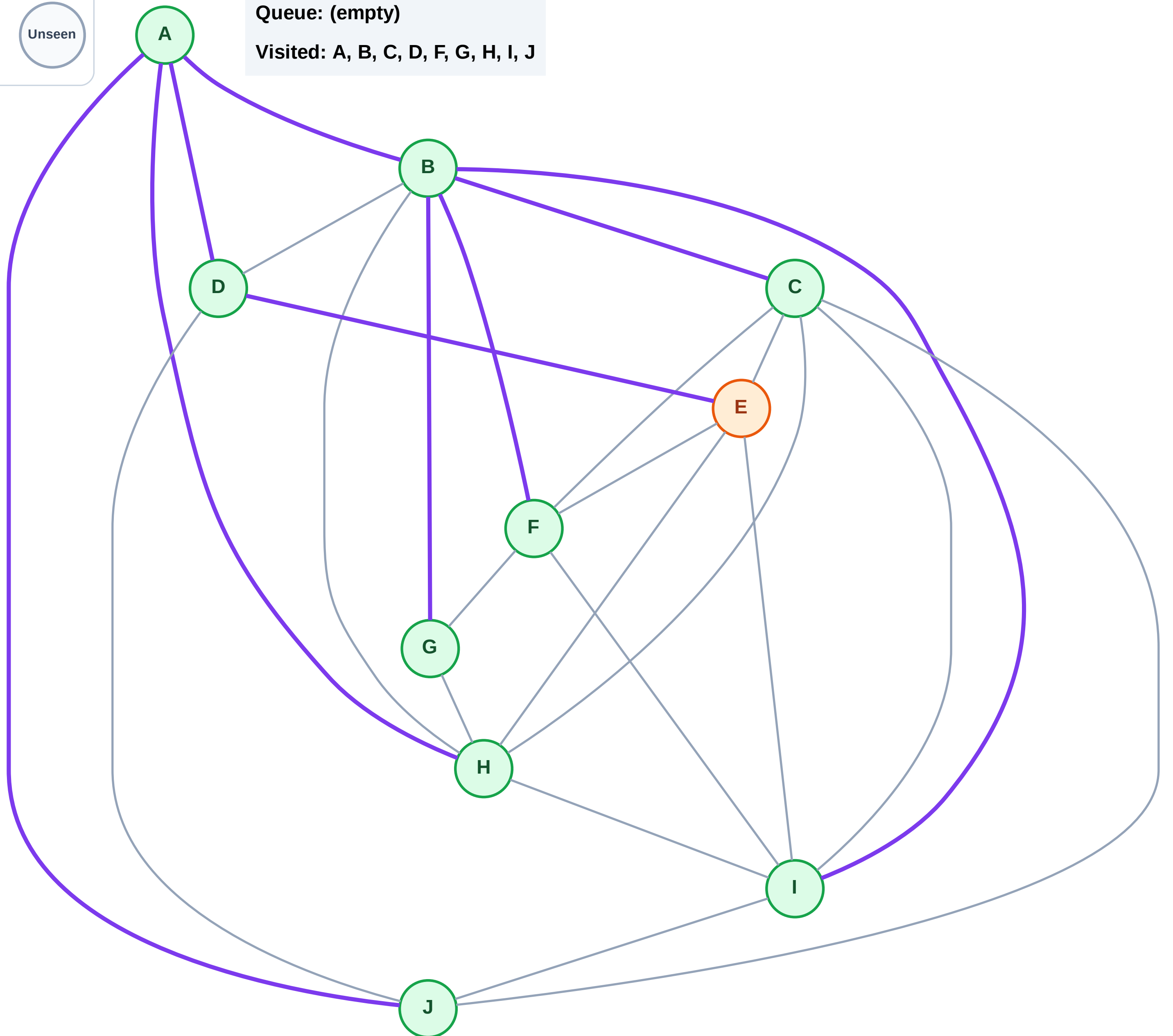
Step 20: Process node E

Legend



Queue: (empty)

Visited: A, B, C, D, F, G, H, I, J



BFS Traversal

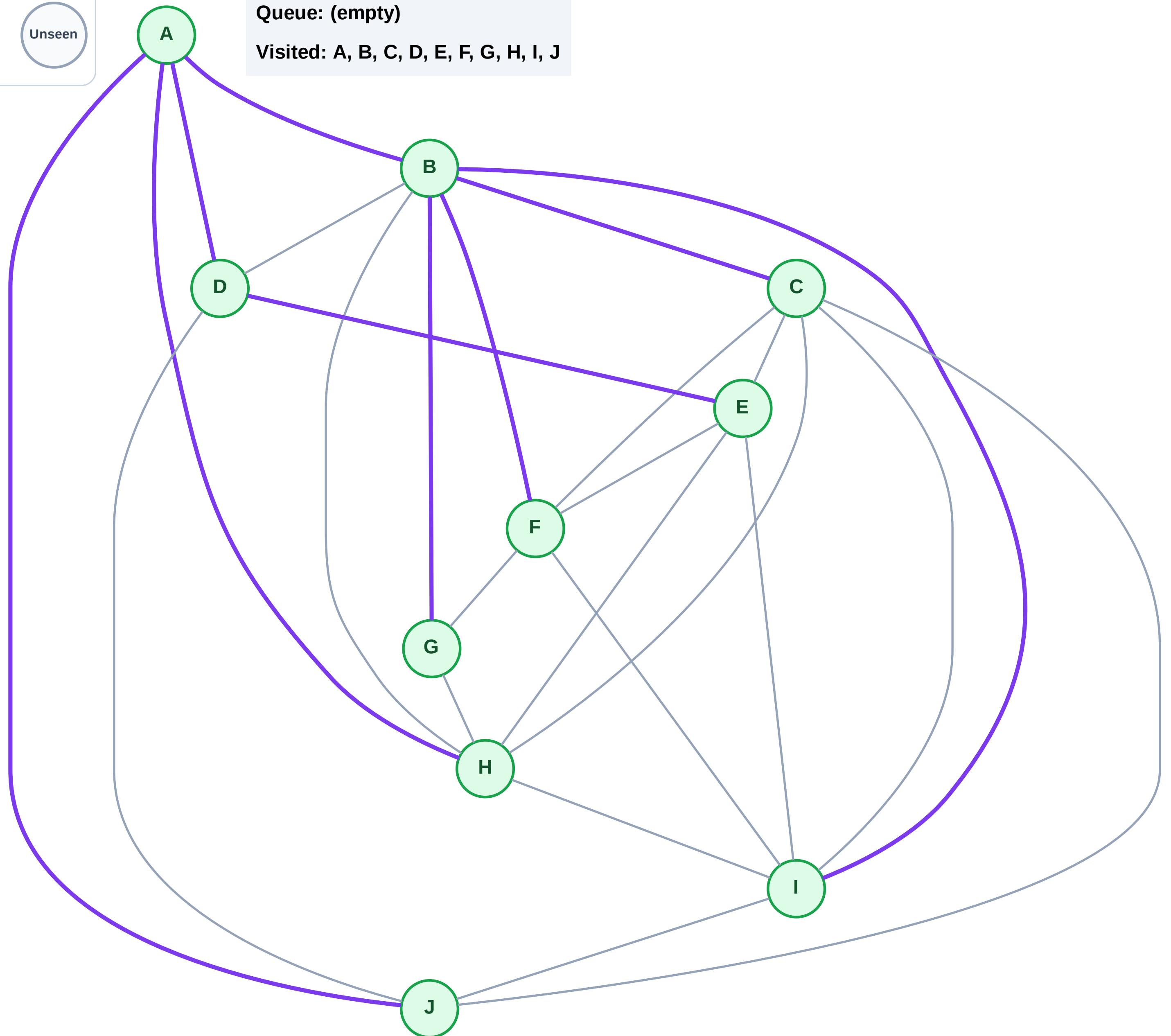
Step 21: No new neighbors from E

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

Legend

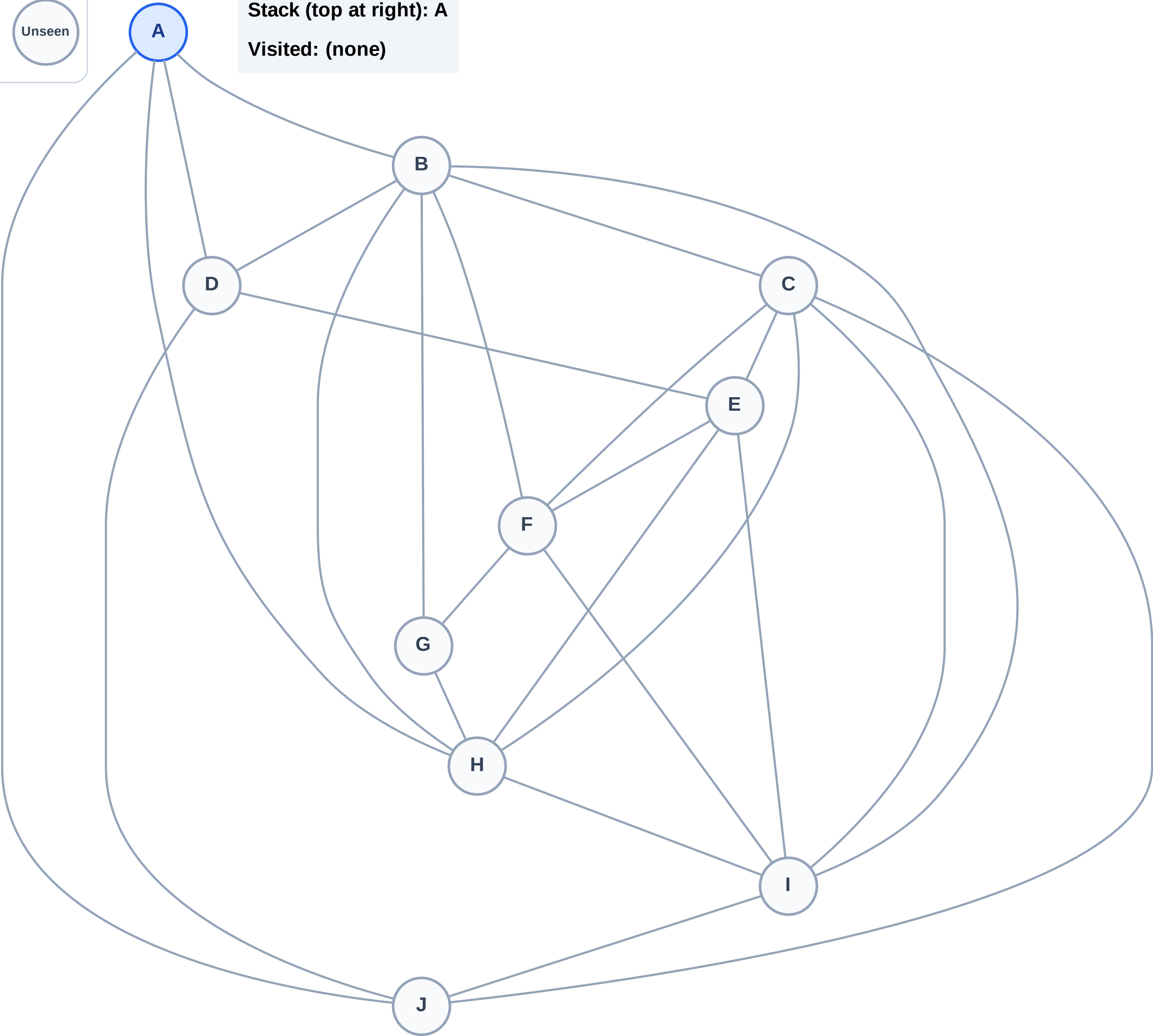
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

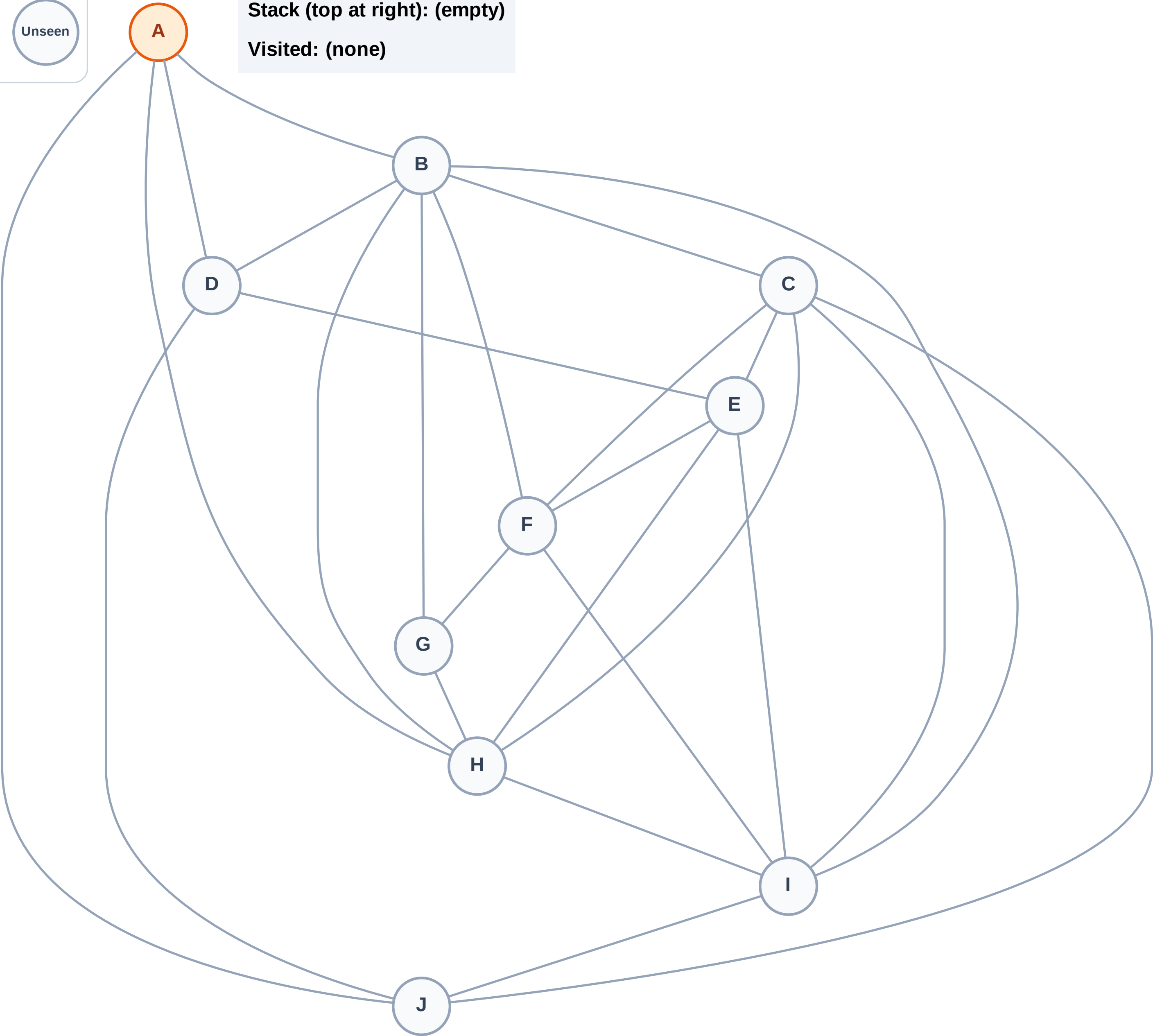
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



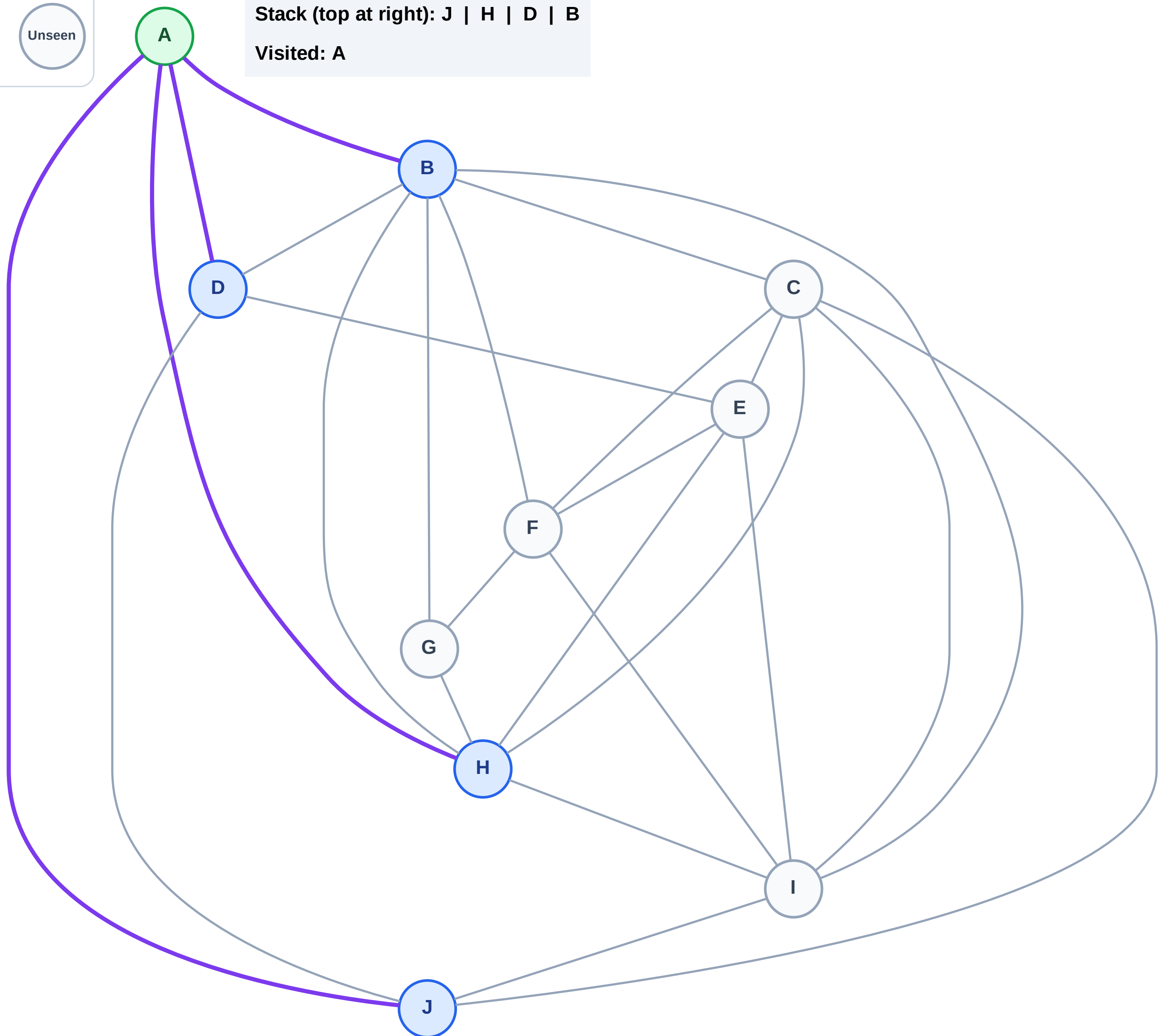
DFS Traversal

Step 3: Discover J, H, D, B from A

Stack (top at right): J | H | D | B

Visited: A

Legend



DFS Traversal

Step 4: Process node B

Legend

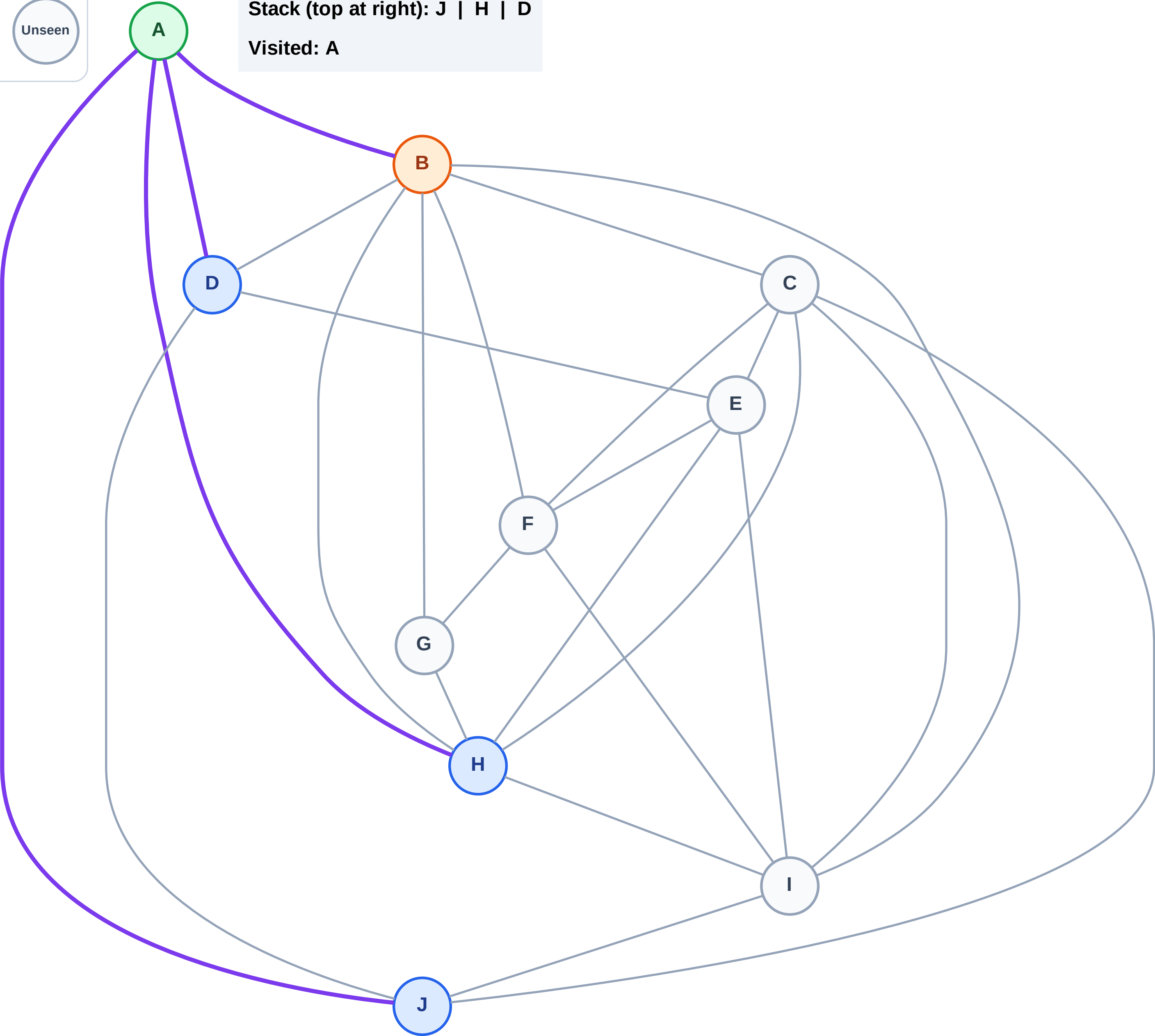
Complete

Processing

Discovered

Unseen

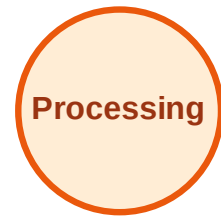
Stack (top at right): J | H | D
Visited: A



DFS Traversal

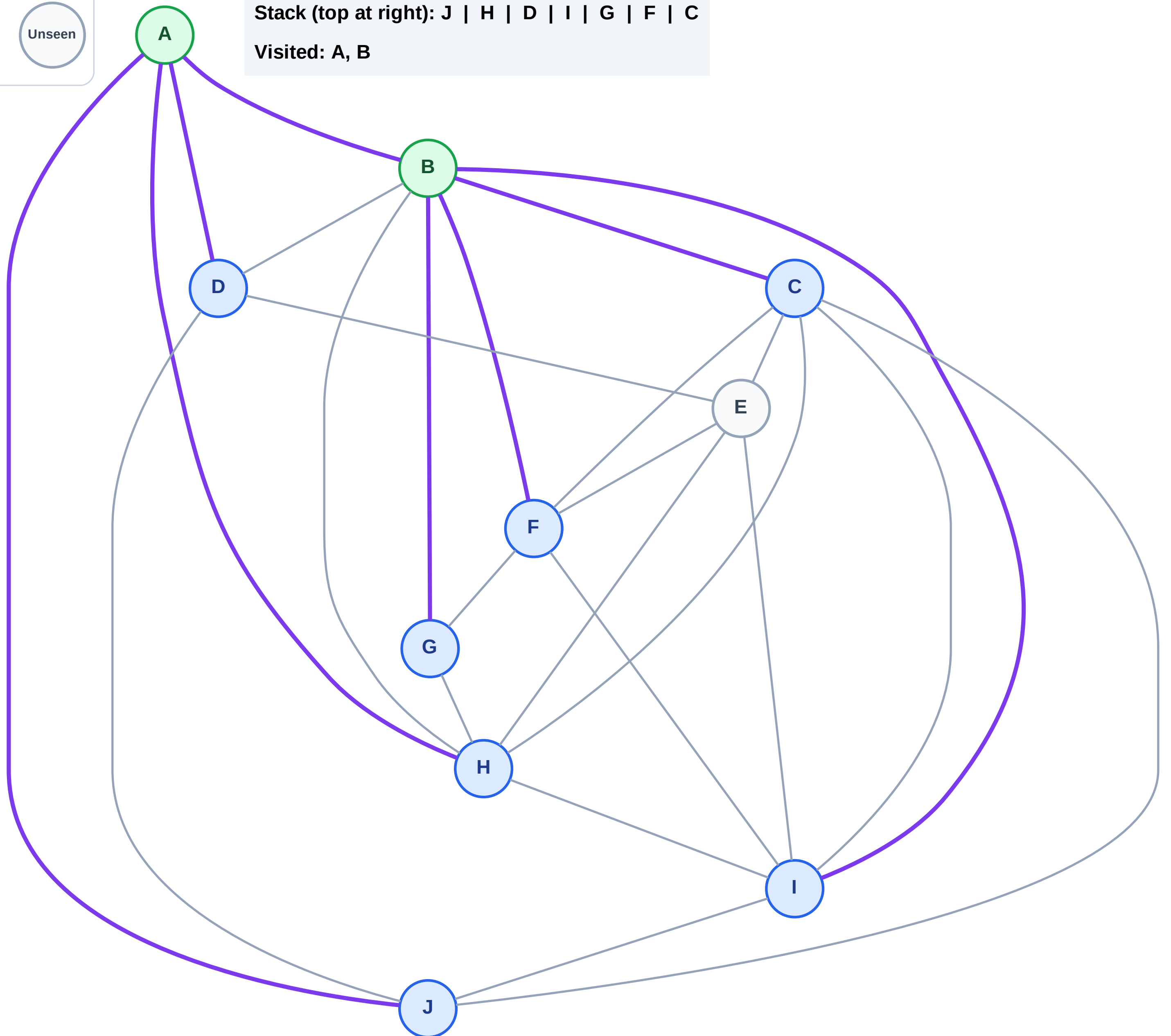
Step 5: Discover I, G, F, C from B

Legend



Stack (top at right): J | H | D | I | G | F | C

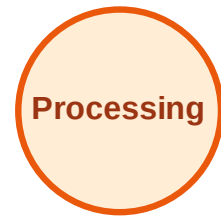
Visited: A, B



DFS Traversal

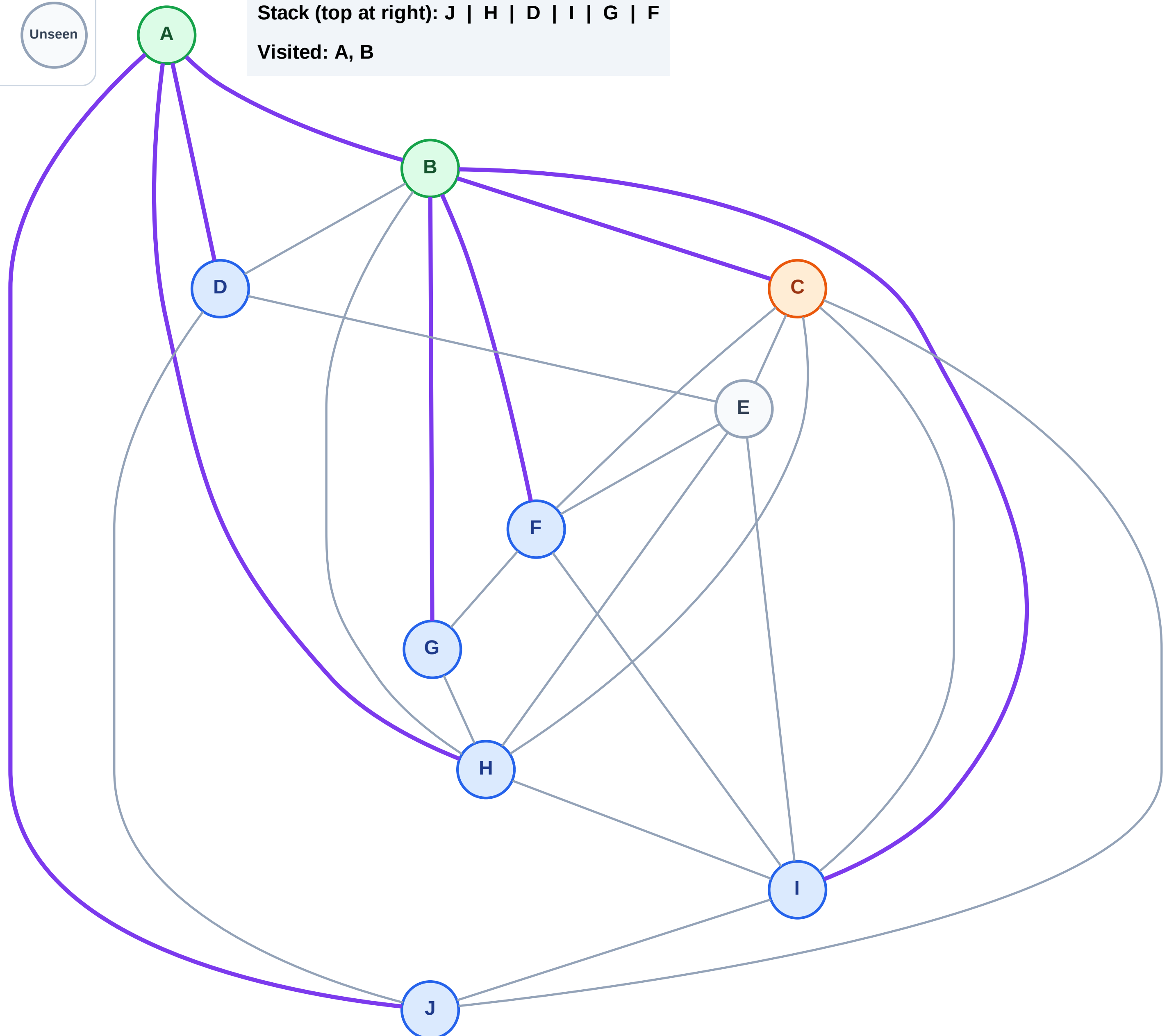
Step 6: Process node C

Legend



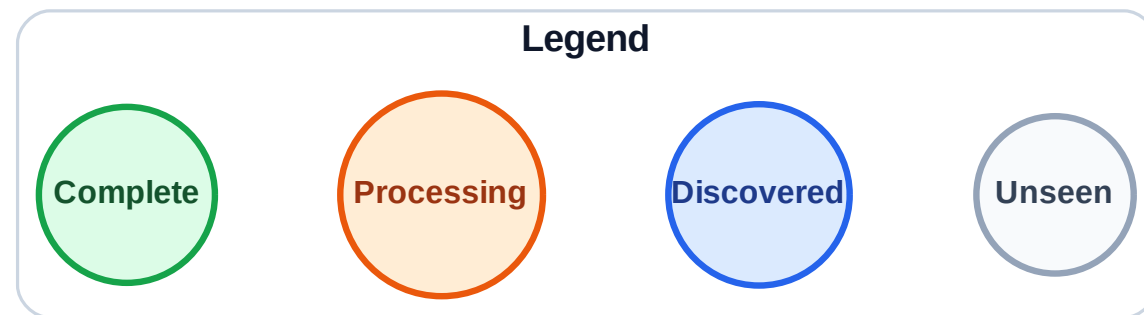
Stack (top at right): J | H | D | I | G | F

Visited: A, B

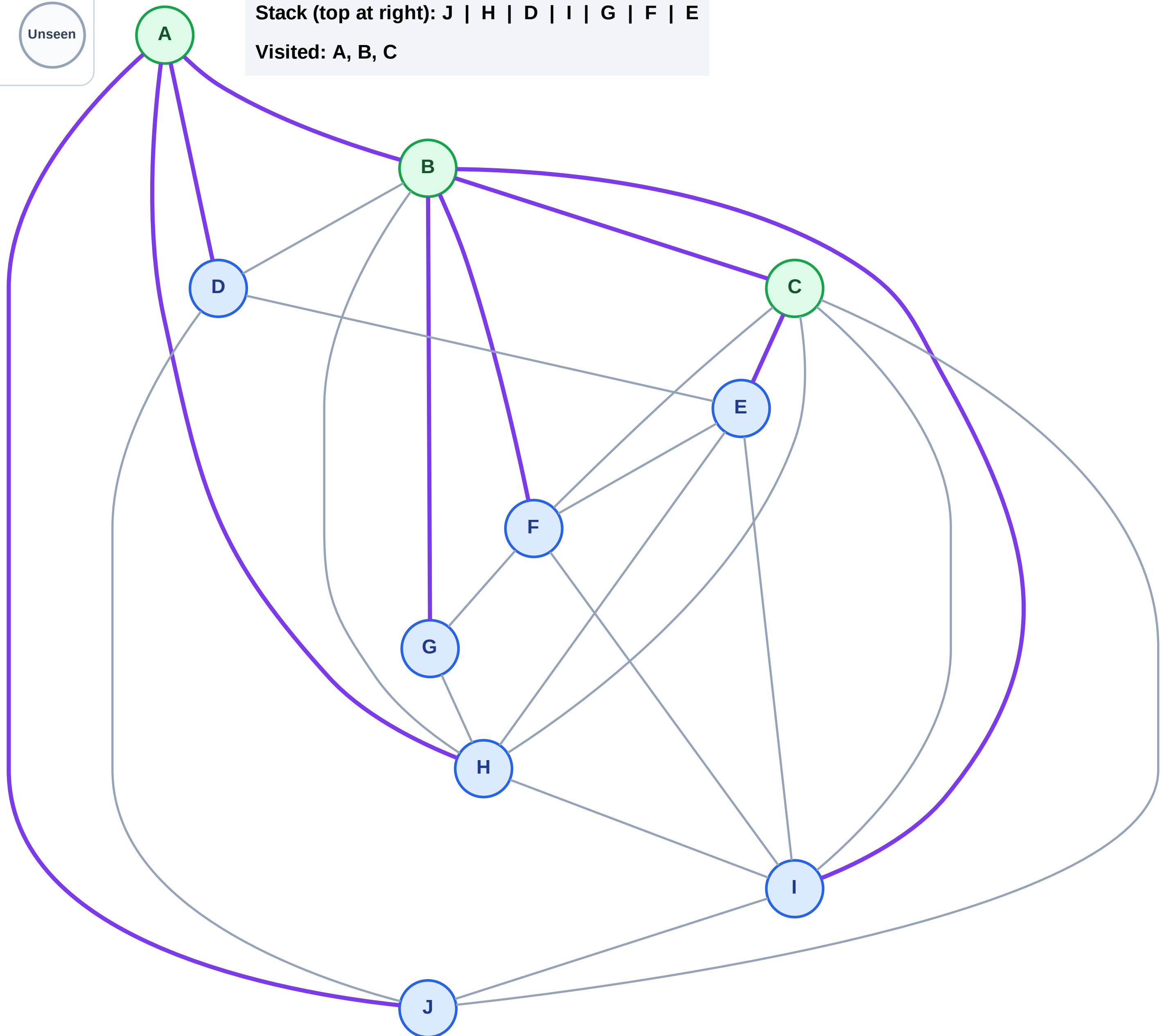


DFS Traversal

Step 7: Discover E from C



Stack (top at right): J | H | D | I | G | F | E
Visited: A, B, C



DFS Traversal

Step 8: Process node E

Legend

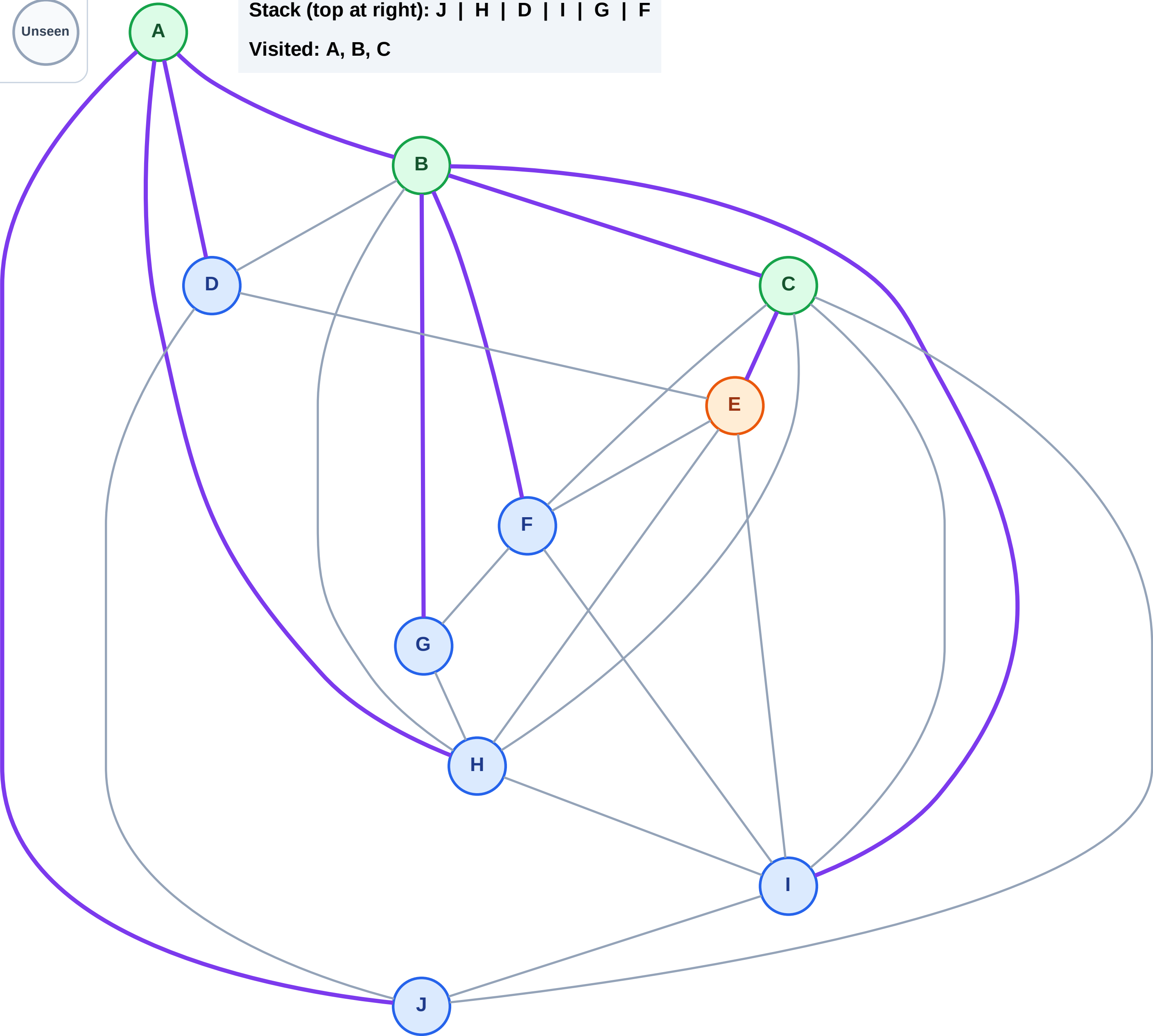
Complete

Processing

Discovered

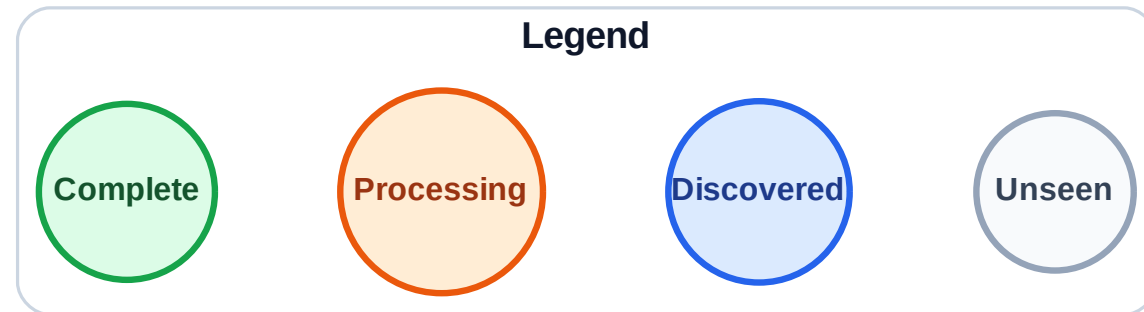
Unseen

Stack (top at right): J | H | D | I | G | F
Visited: A, B, C

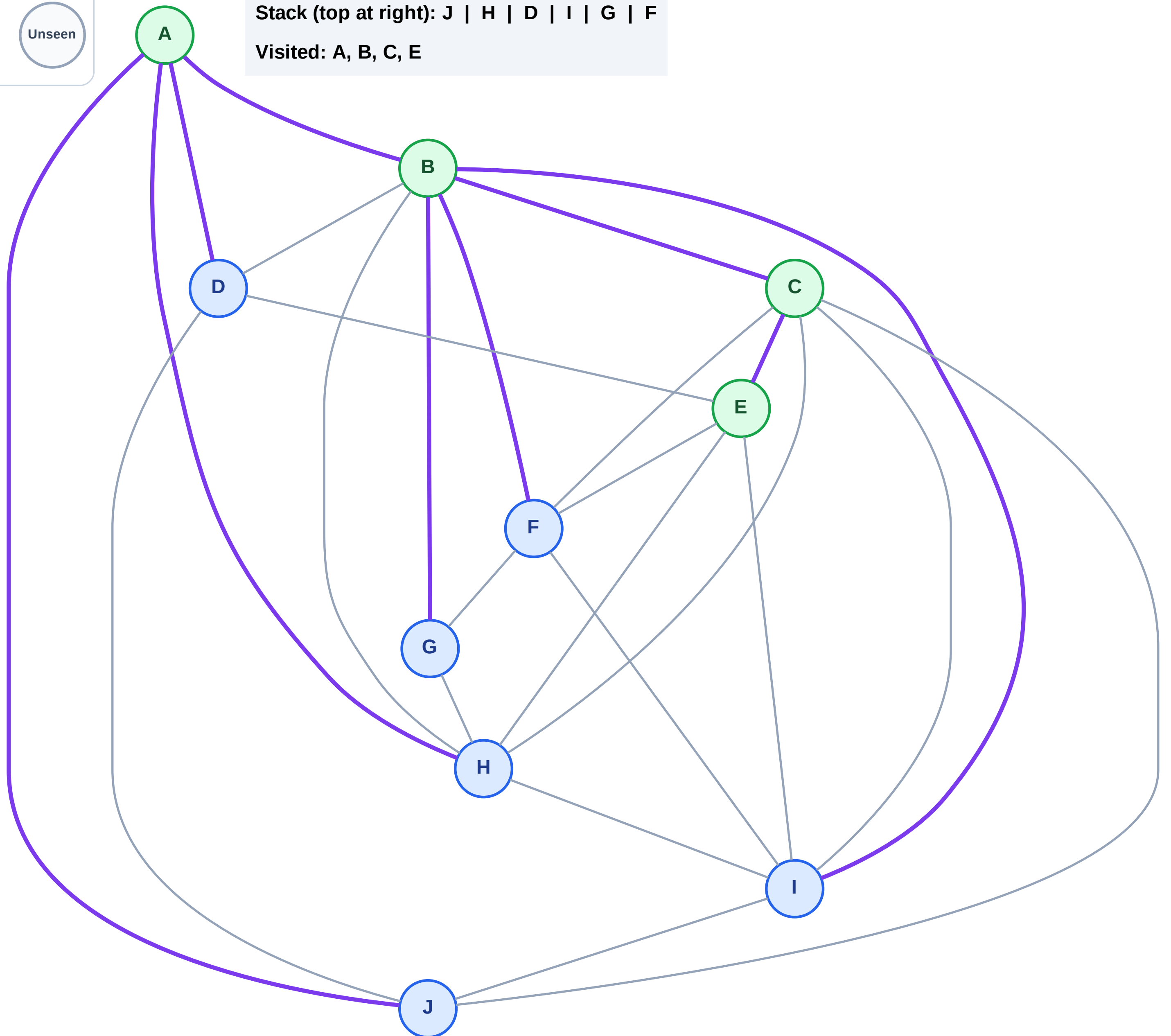


DFS Traversal

Step 9: No new neighbors from E

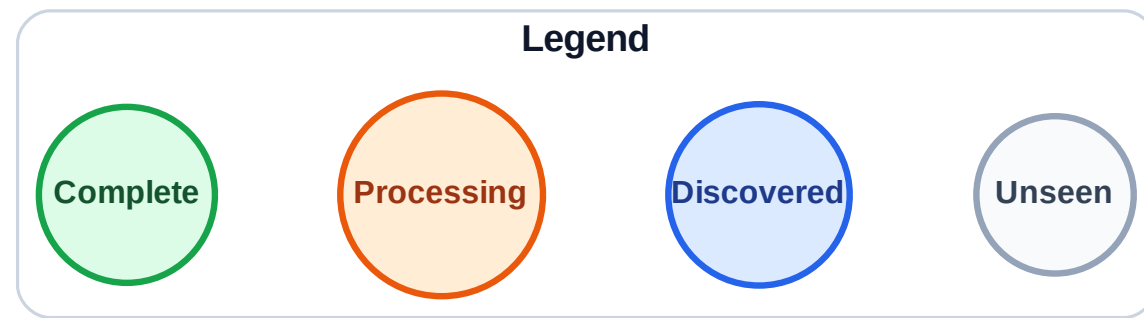


Stack (top at right): J | H | D | I | G | F
Visited: A, B, C, E

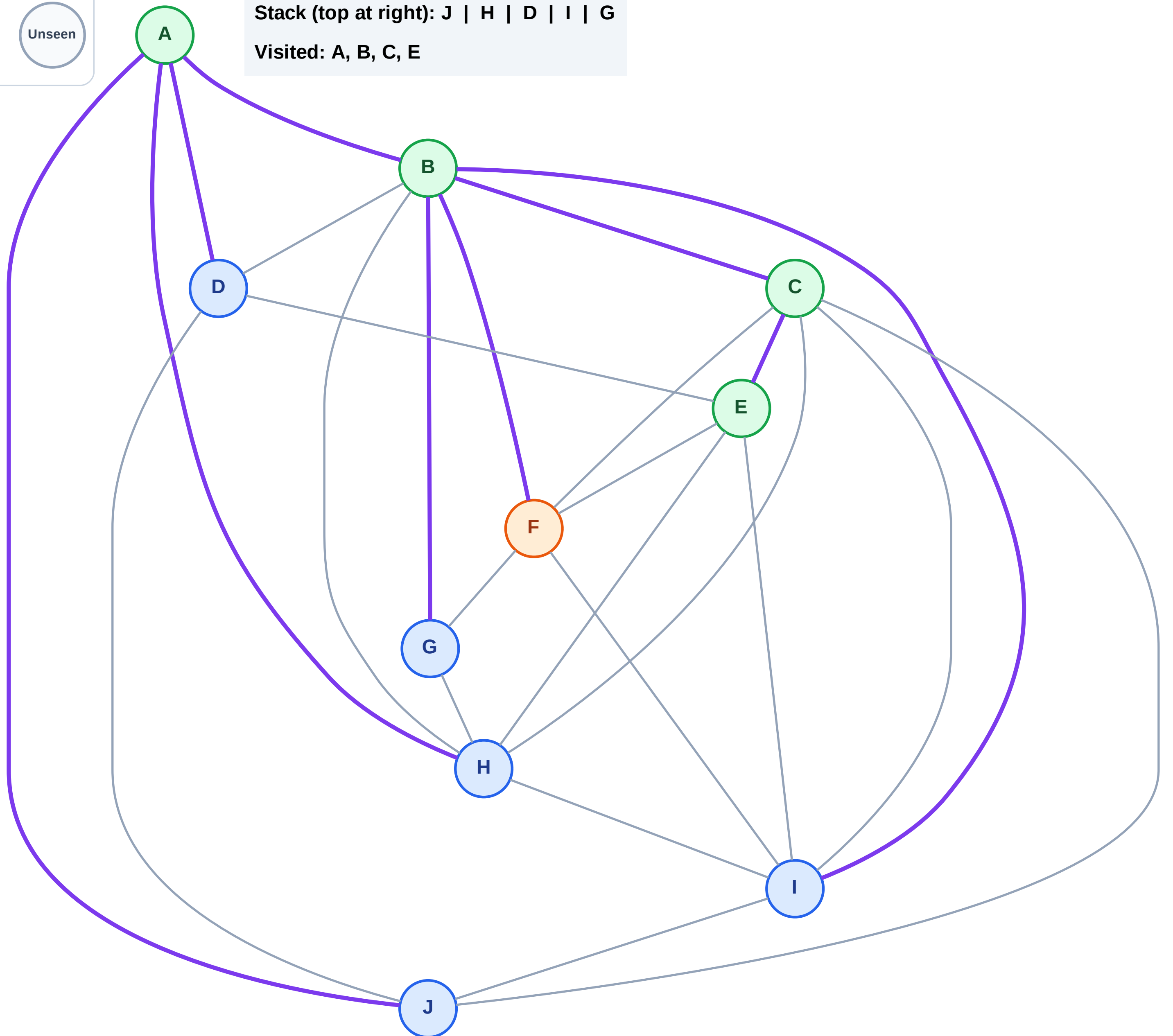


DFS Traversal

Step 10: Process node F

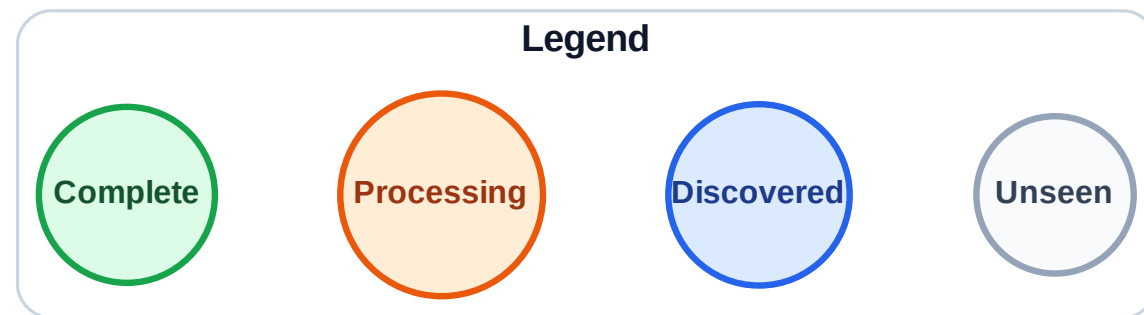


Stack (top at right): J | H | D | I | G
Visited: A, B, C, E



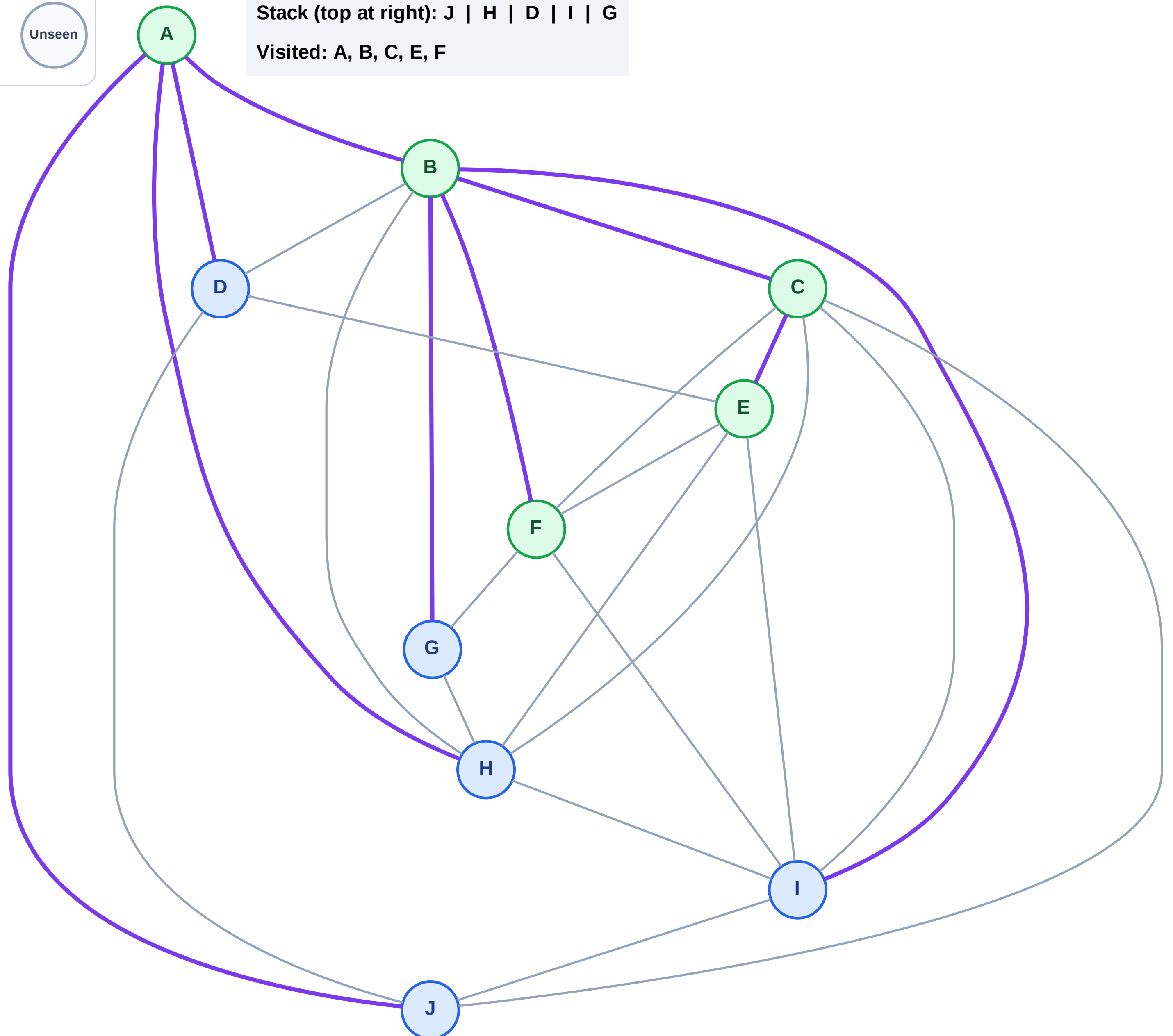
DFS Traversal

Step 11: No new neighbors from F



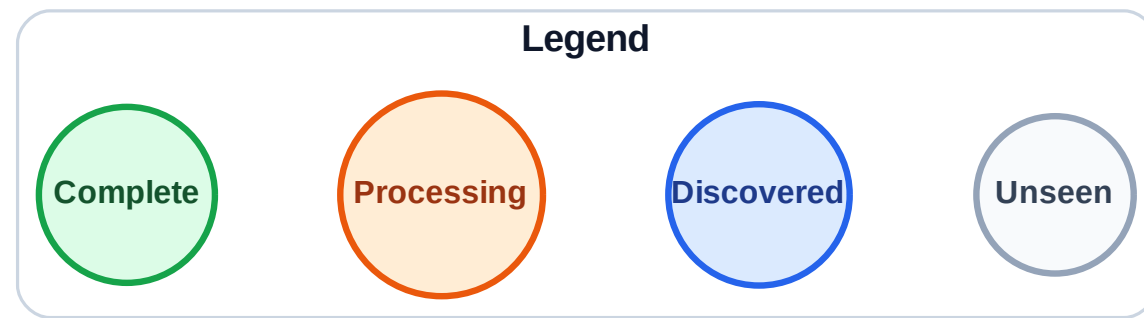
Stack (top at right): J | H | D | I | G

Visited: A, B, C, E, F

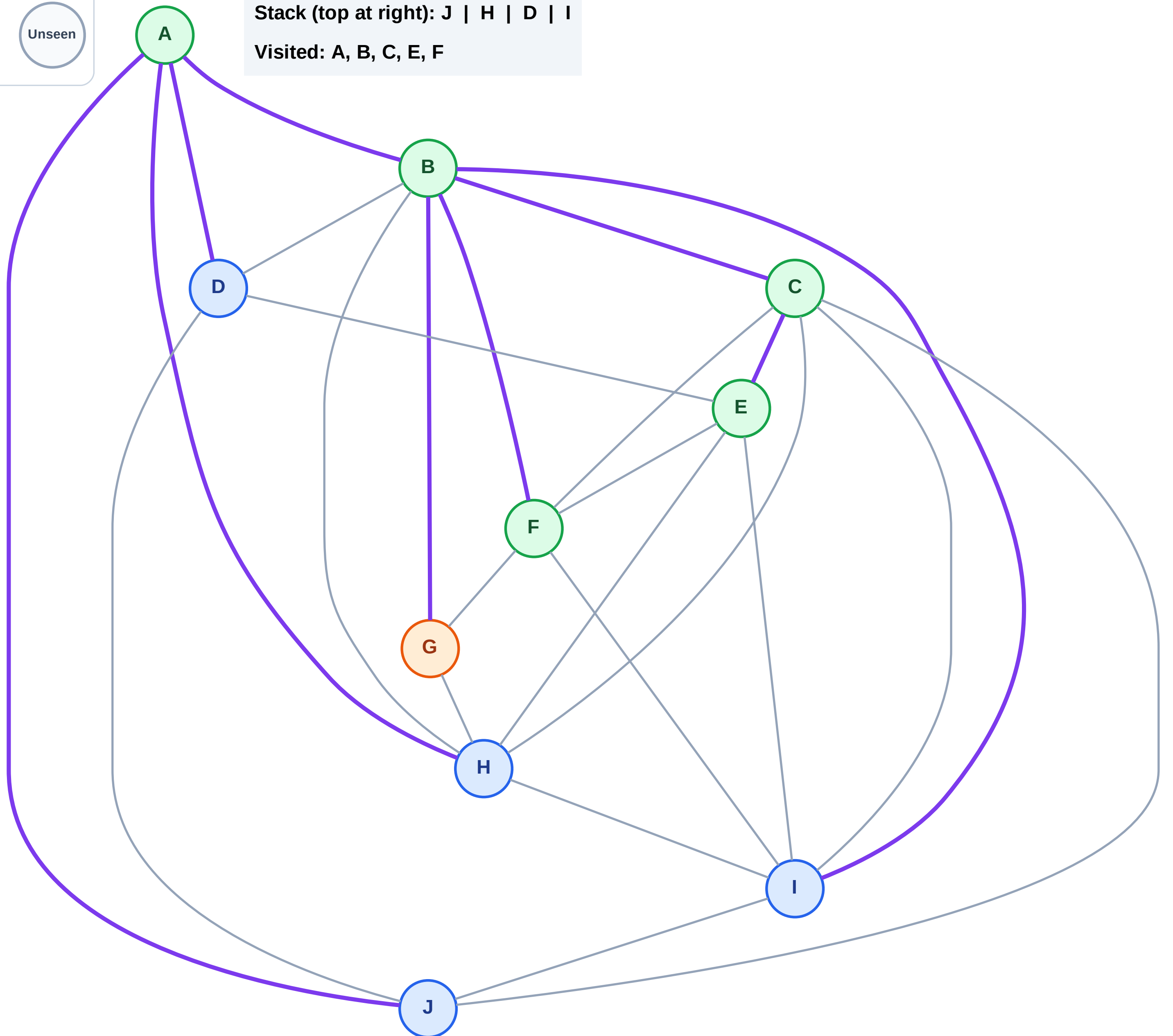


DFS Traversal

Step 12: Process node G



Stack (top at right): J | H | D | I
Visited: A, B, C, E, F

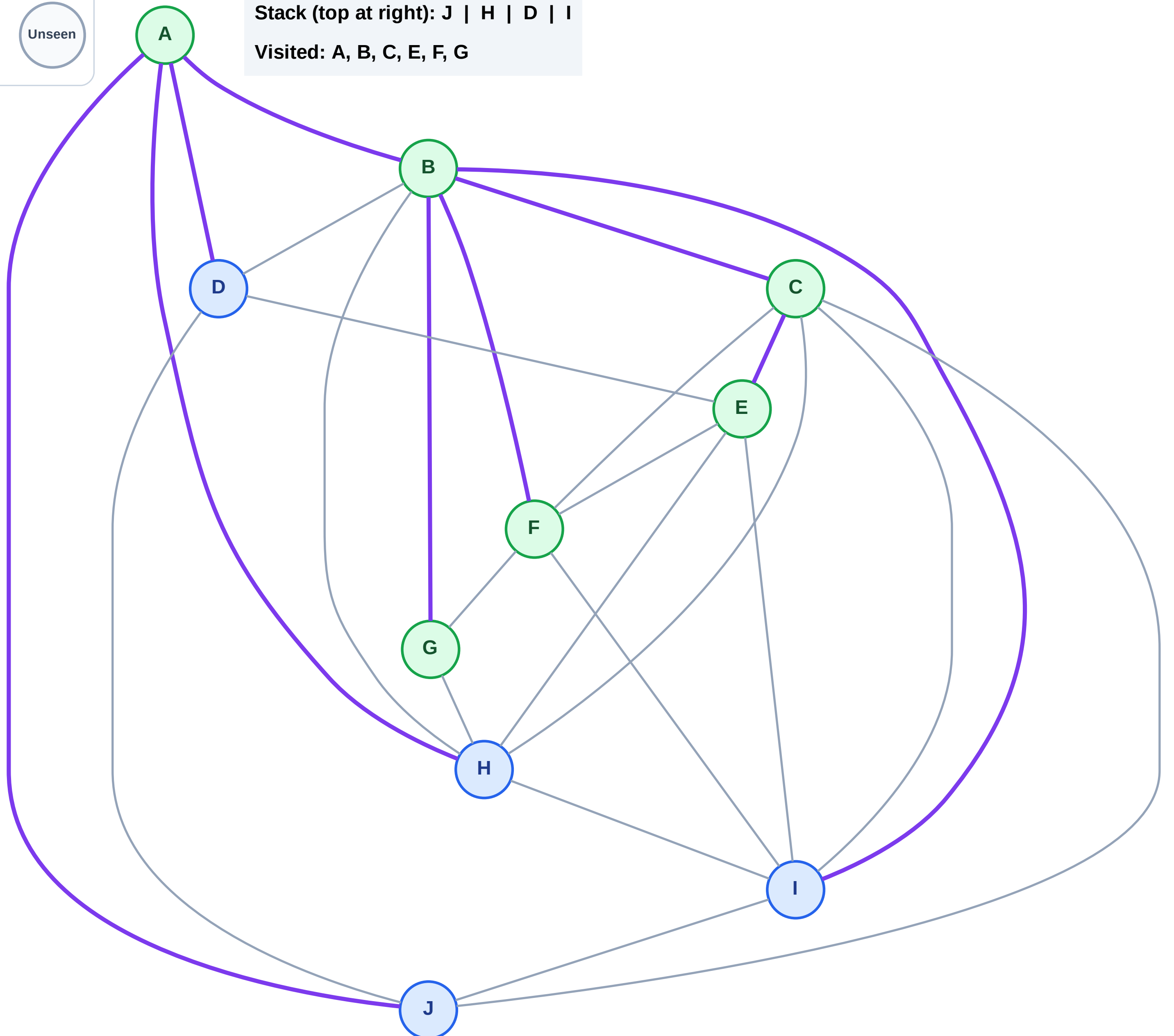
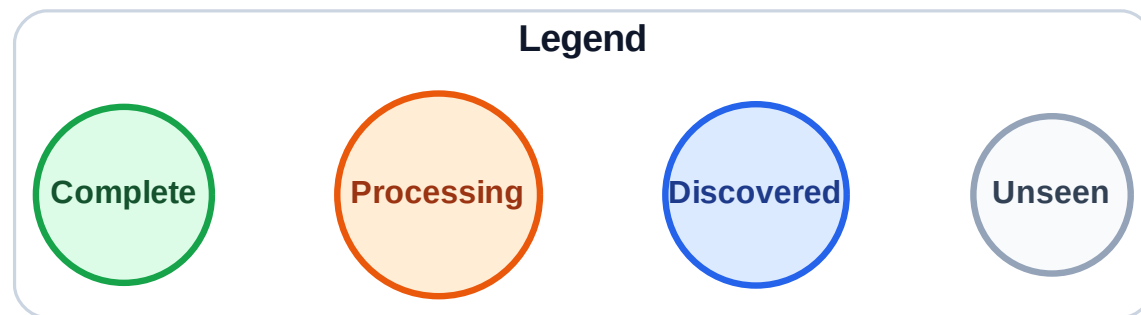


DFS Traversal

Step 13: No new neighbors from G

Stack (top at right): J | H | D | I

Visited: A, B, C, E, F, G



DFS Traversal

Step 14: Process node I

Legend

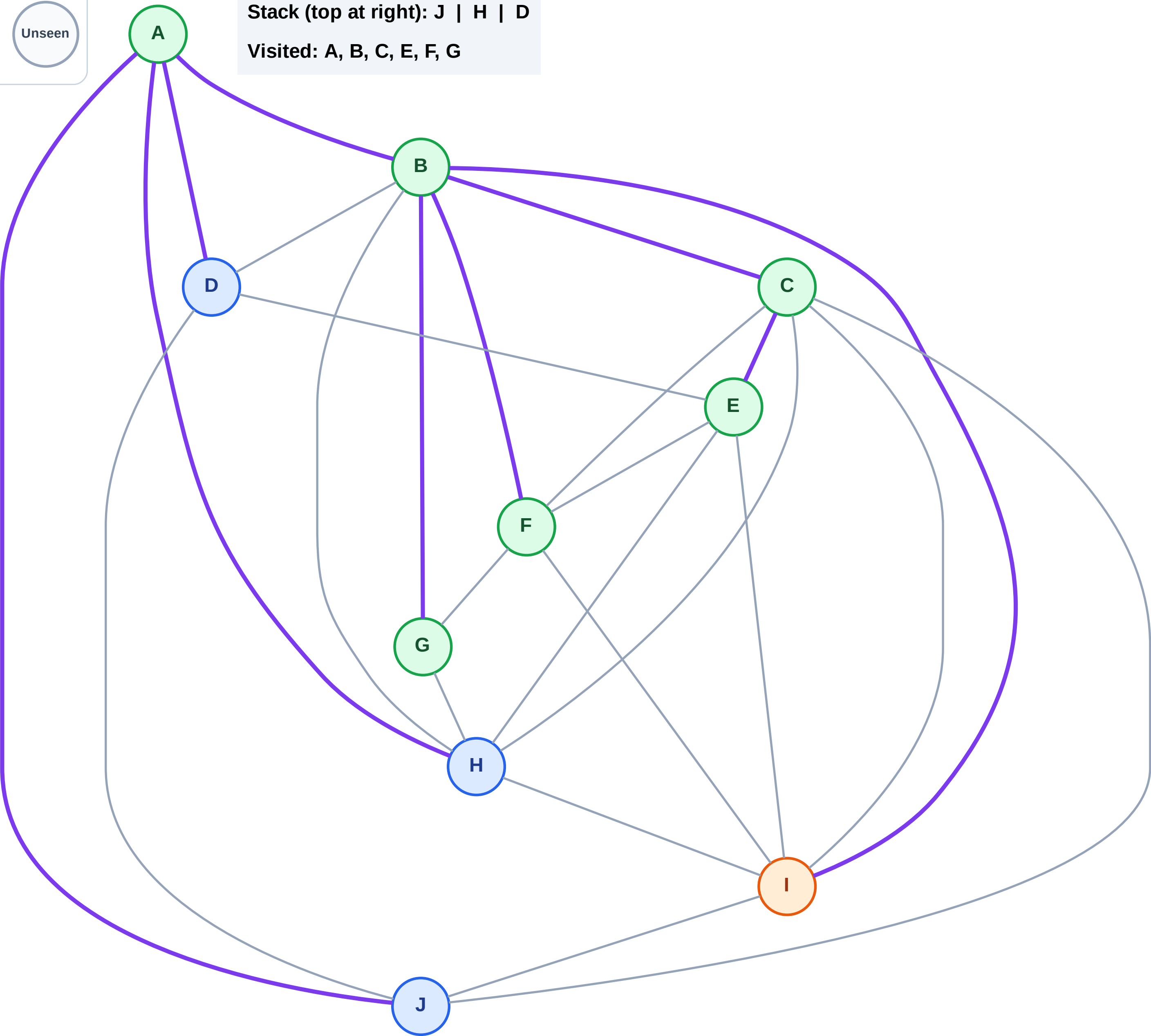
Complete

Processing

Discovered

Unseen

Stack (top at right): J | H | D
Visited: A, B, C, E, F, G



DFS Traversal

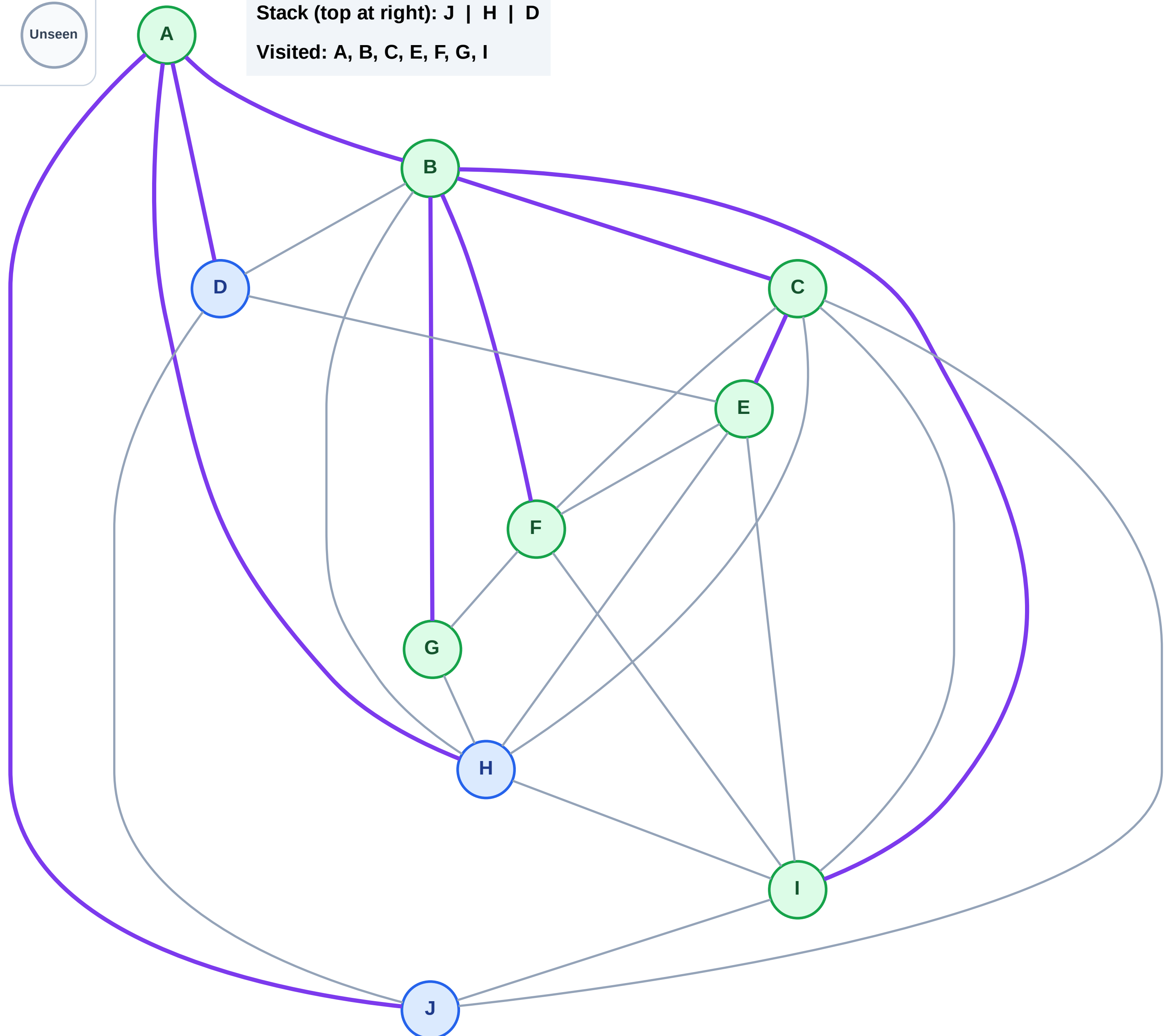
Step 15: No new neighbors from I

Legend



Stack (top at right): J | H | D

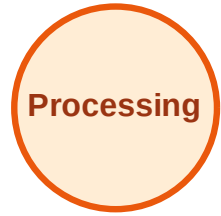
Visited: A, B, C, E, F, G, I



DFS Traversal

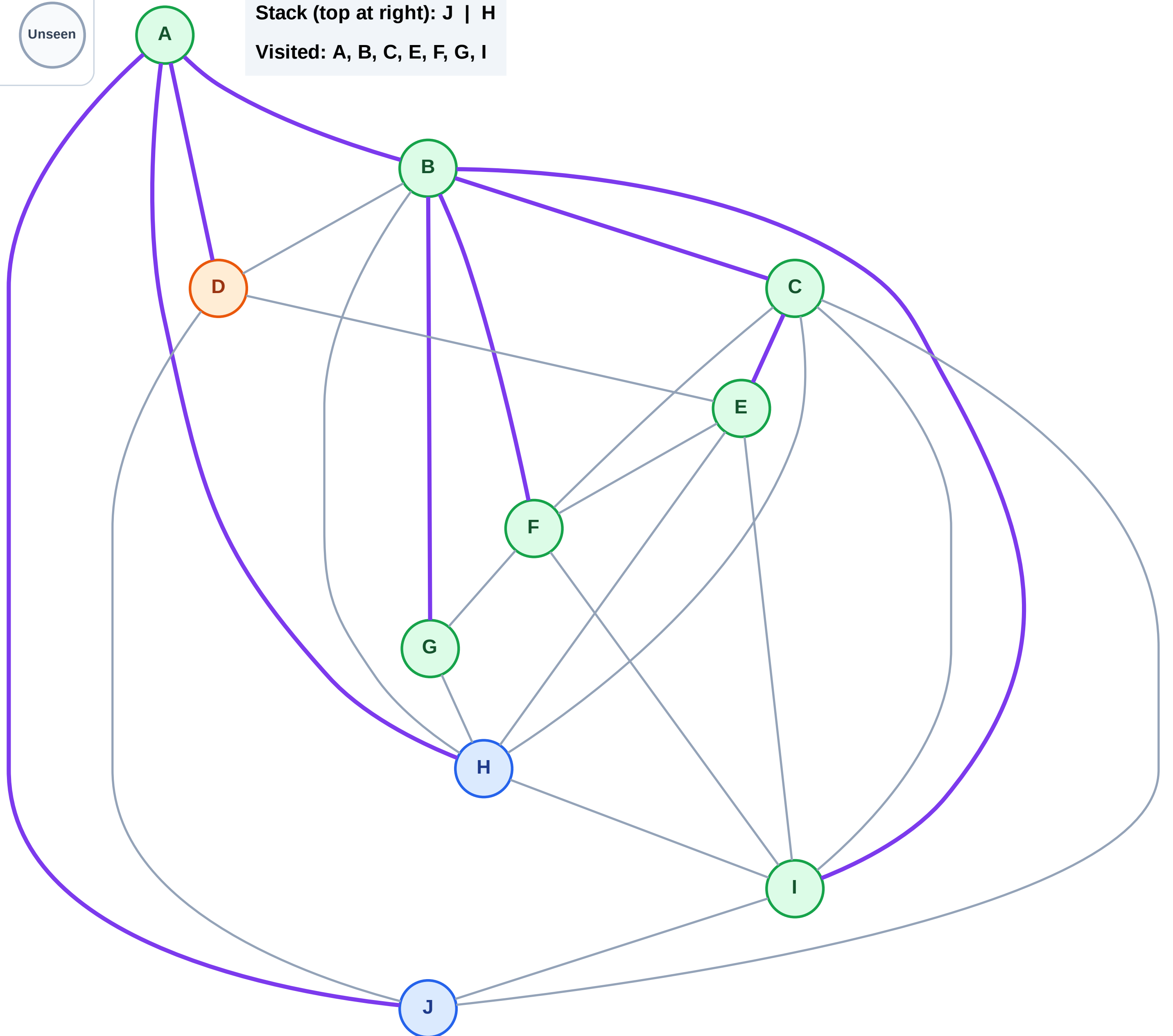
Step 16: Process node D

Legend



Stack (top at right): J | H

Visited: A, B, C, E, F, G, I



DFS Traversal

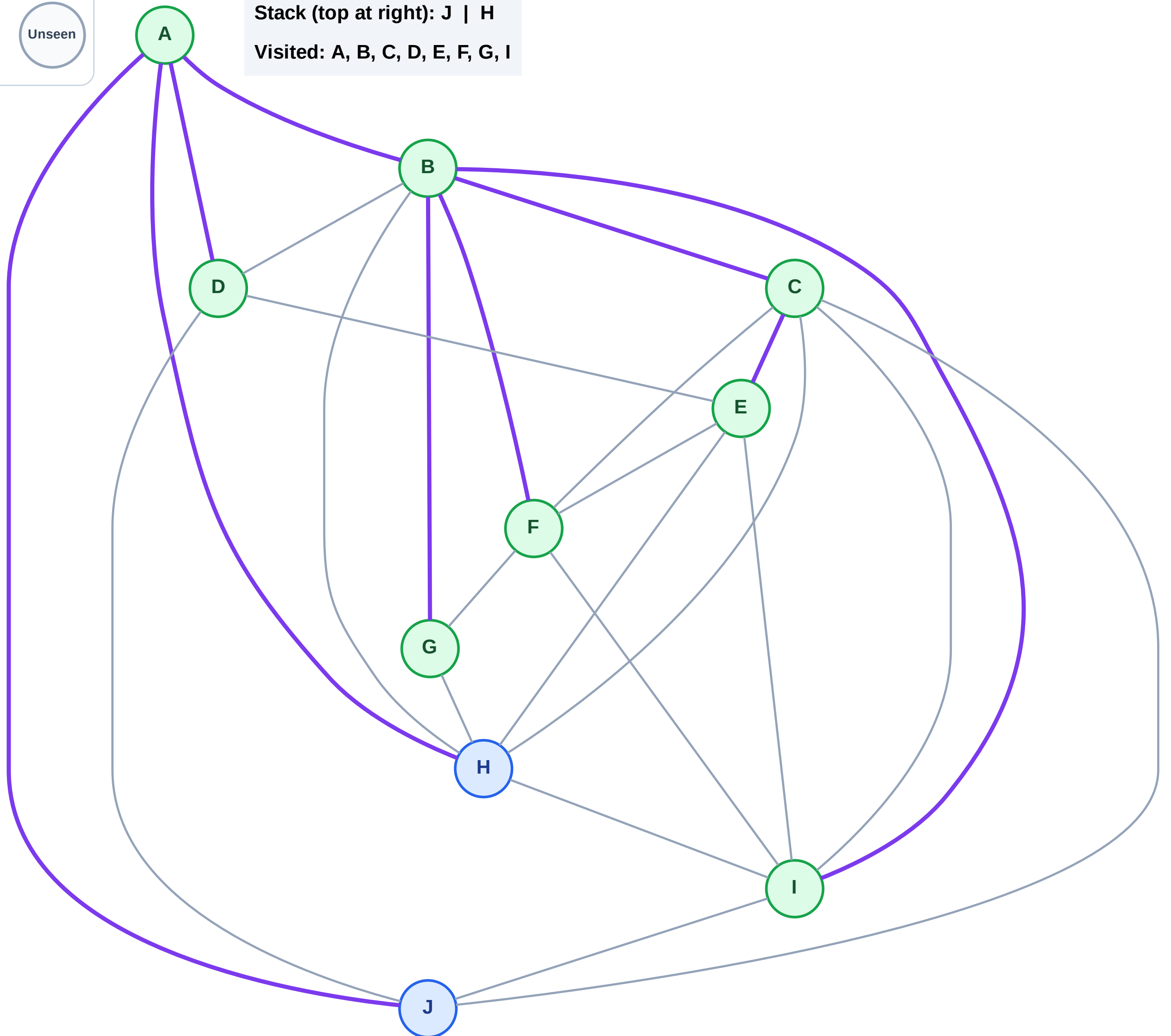
Step 17: No new neighbors from D

Legend



Stack (top at right): J | H

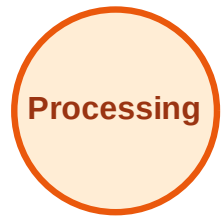
Visited: A, B, C, D, E, F, G, I



DFS Traversal

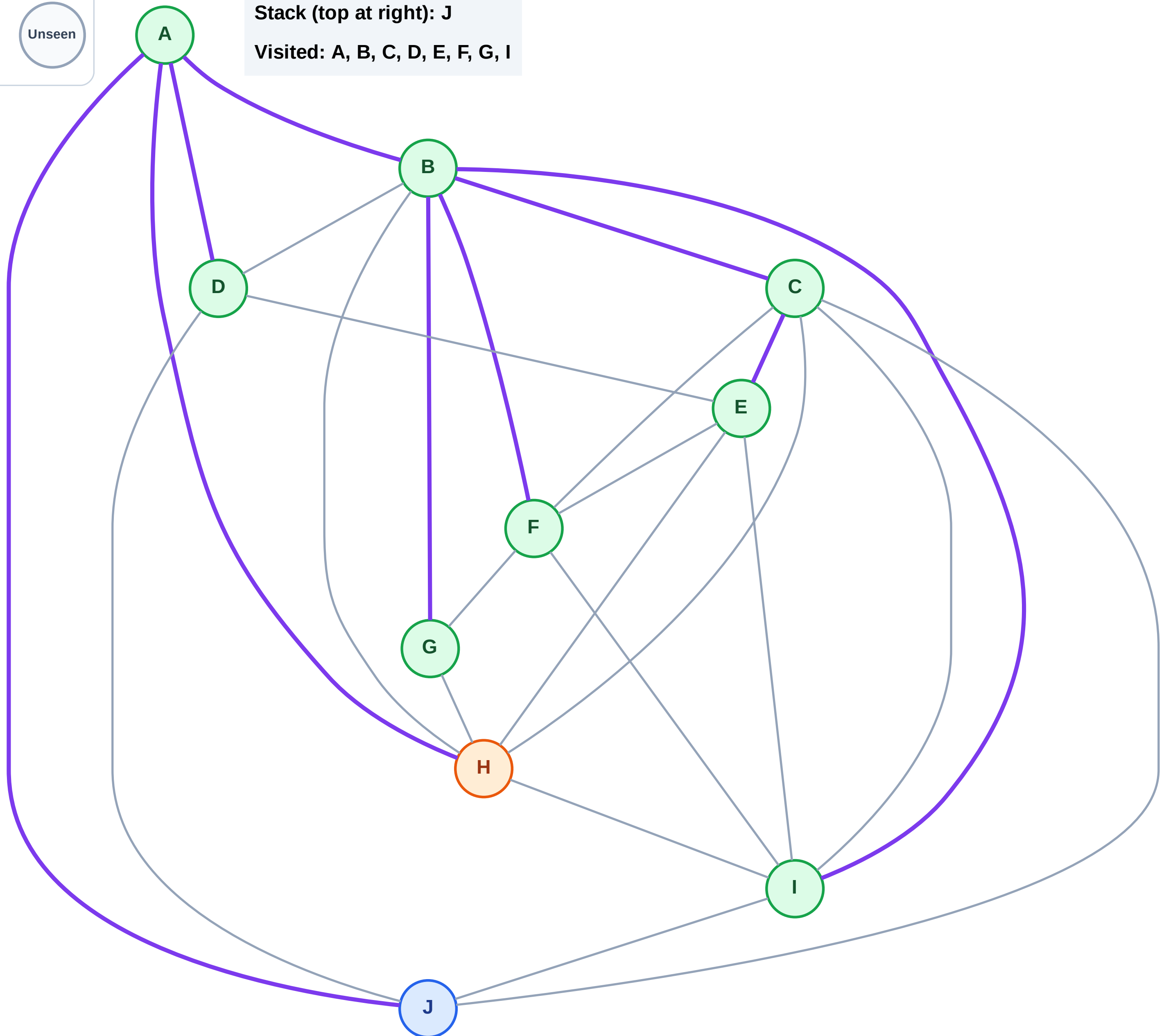
Step 18: Process node H

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, I



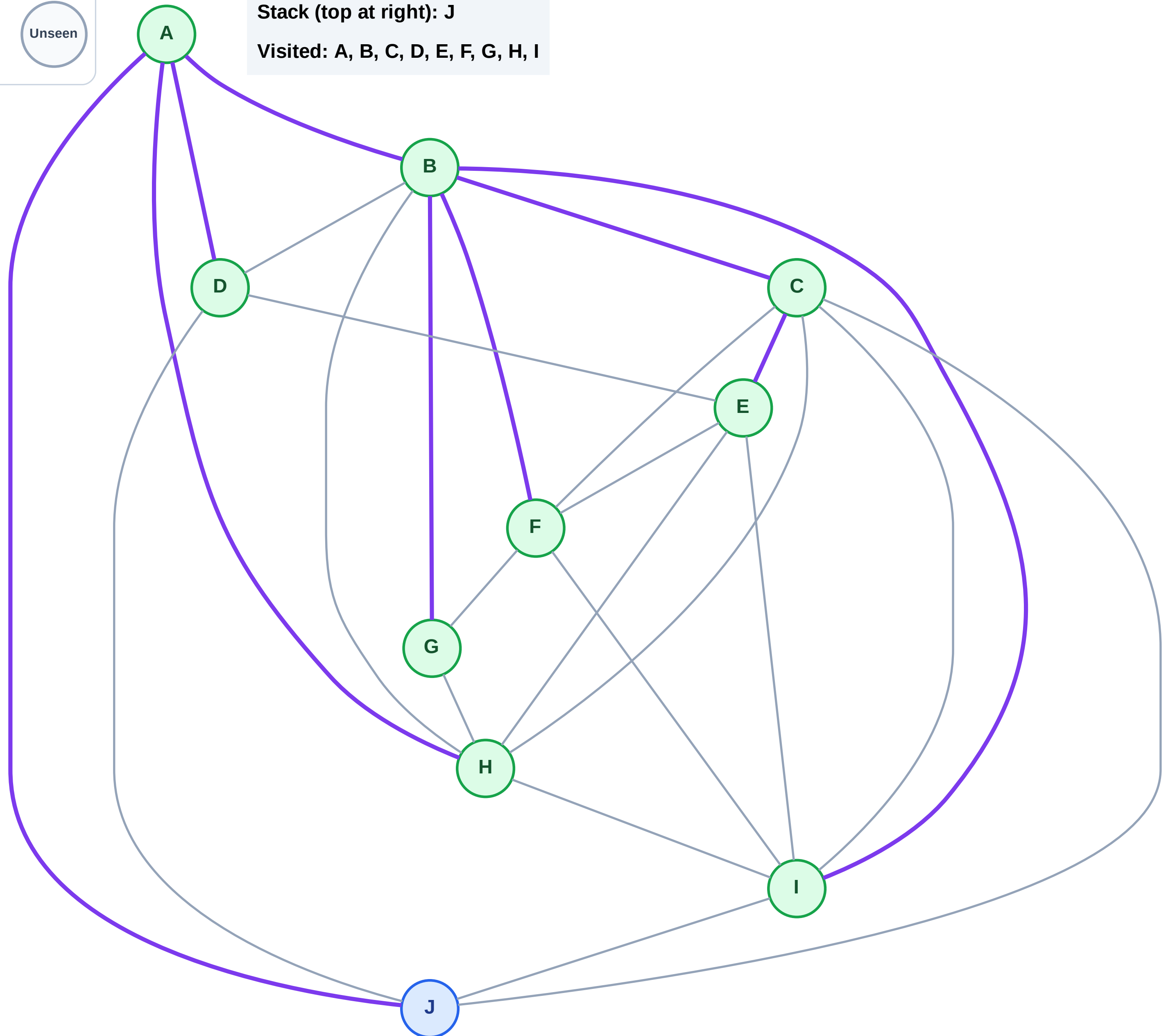
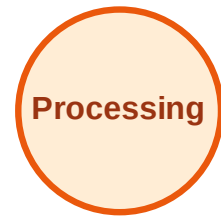
DFS Traversal

Step 19: No new neighbors from H

Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I

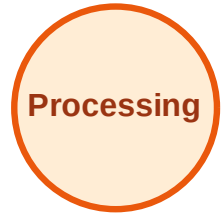
Legend



DFS Traversal

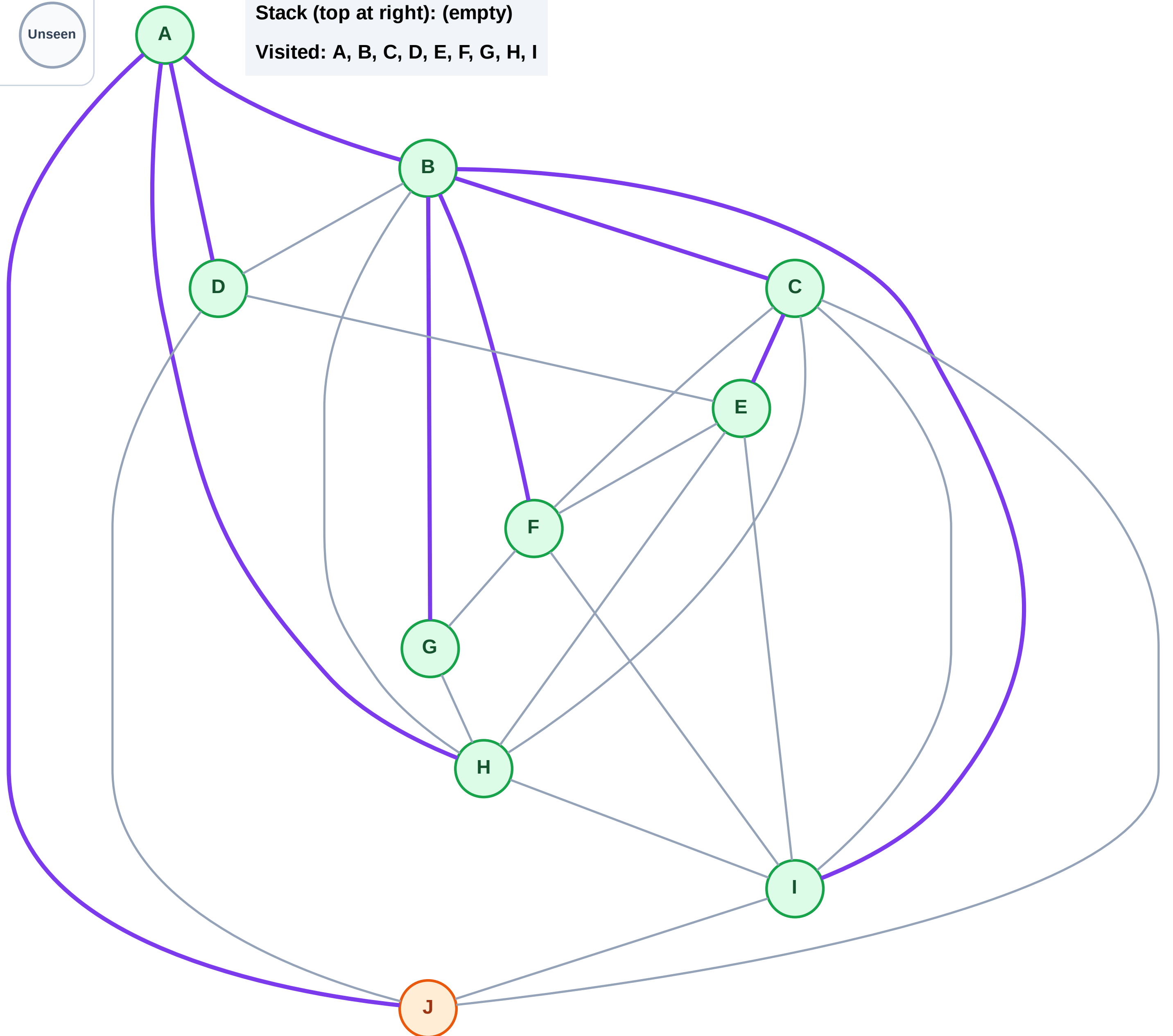
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

